T106/B126 - COMPUTER PROGRAMMING B.TECH 1ST YEAR

Unit-I (Two Marks)

1. **First calculating device & First calculator of the world?**

   Abacus, Pascaline.

2. **Mechanical calculator & First Mechanical computer?**

   Pascaline, Analytical Engine.

3. **First electro-mechanical computer & First Electronic computer and when it was built?**

   ASCC - Automatic Sequence Controlled Computer, **ENIAC** - Electronic Numerical Integrator and Computer built in 1943.

4. **Who invented the Napier's bones?**

   Napier's bones was built by John Napier, a Scotland mathematician in 1617.

5. **Who invented the Pascaline and when?**

   Pascaline was invented by Blaise Pascal, a French mathematician in 1642 AD.

6. **Who invented the Slide Rule?**

   Sliderule was invented by William Oughtred, an English mathematician in 1621.

7. **Who invented the Punched card system?**

   Punched card system was invented by Joseph Marie Jacquard, a French textile manufacturer in 1801.

8. **Who invented the fast counting machine and what its name?**

   Herman Hollerith, an American mathematician invented the fast counting machine and named it as Tabulating Machine.

9. **Who designed the Difference Engine and for what purpose?**

   Charles Babbage, an English mathematician designed the Difference Engine in 1822 to calculate mathematical tables.

10. **Define Computer?**

    A Computer is an electronic device which stores and processes information or data to give meaningful results. Processing is done with the help of instructions given by the user. The data, as well as instructions provided to the computer are known as INPUT and the processed result obtained at the end is known as OUTPUT.

11. **What is an Information Technology?**

    Information Technology is technology that merges computing with high-speed communication links carrying data, sound and video.

12. **Define Data, Information**.

    Data consists of the raw facts and figures that are processed into information.

    Information is summarized data or otherwise manipulated data that is useful for decision making.

13. **What are the major Components of 1st, 2nd, 3rd and 4th generation Computers?**

    The major components of 1st, 2nd, 3rd and 4th generation computers are Vacuum tubes (valves), Transistors, IC's (Integrated Circuits) and VLSI (Very Large Scale Integration).

14. **Expand ASCC, ENIAC and EDSAC?**

    ASCC - Automatic Sequence Controlled Computer

    ENIAC - Electronic Numerical Integrator and Computer

    EDSAC – Electronic Delay Storage Automatic Calculator

15. **What is a Personal Computer? List the different types of PCs?**

    Personal Computers (PCs) are desktop, tower or portable computers that can run easy-to-use programs such as word processing or spreadsheets.

    PCs include desktop, floorstanding units, laptops, notebooks, subnotebooks, pocket PCs and pen computers.

16. **What is a Mainframe computer? Where are they used?**

    Mainframes are traditional size of computer and are used in large companies to handle millions of transactions occupying specially wired, air-conditioned rooms and capable of great processing speeds and data storage. Mainframes are used in large organizations for large scale jobs such as banks, insurance companies and airlines.

17. **Define CPU.**

    The brain of any computer system is the Central Processing Unit. This unit controls the entire computer and executes the given instructions. CPU consists of three major sections and they are,

    a. Primary storage section or Memory unit.
    b. Arithmetic and logical unit (ALU) and
    c. Control unit.

18. **Define Control Unit and ALU.**

    Control Unit: The control unit acts as a central nervous system for the other components of the computer. It controls and directs the operation of the devices and the entire system. Control unit performs two main functions

    i. Supervises transfer of data.
    ii. Selects and translates or interprets instructions.

    ALU: All calculations are performed and all comparisons are made in the arithmetic and logical unit. In this unit only all the processing takes place.

19. **List out the characteristics of Computer.**

    Speed, Accuracy, Diligence, Reliability, Storage Capability, Versatility, Resource Sharing.

20. **Define Hardware and Software.**

    Hardware: All physical components of a computer system are called hardware. We can see, touch and feel the hardware components. CPU, keyboard, printer, monitor, hard-disk drive, input-output devices etc are the hardware of a computer system.

Software is a computer program or a set of instructions required for running a computer system. It is responsible for controlling, integrating and managing the hardware components of a computer and to accomplish specific tasks.

21. What are the types of Software?

Software can be categorized into System software and Application software

System software – Operating system, Device drivers, System utilities, Language Translators.

Application software – DataBase Management System (DBMS), Presentation application, Spreadsheet, Image editor, Word processor, Desktop publishing software (DTP).

22. Define Primary Storage and its Components.

Primary memory, also known as main memory stores data and instructions and just ready for immediate processing. Logically, it is an integral component of the CPU but physically, it is a separate part placed on the computer's motherboard. It is limited in size. The primary memory is classified into Random Access Memory (**RAM**) and Read Only Memory (**ROM**).

23. Define Secondary Storage and its types.

Secondary memory, also known as auxiliary memory or external memory is used for storing instructions (Software programs) and data, since main memory is temporary and limited in size. This memory is least expensive and has much larger storage capacity than primary memory. Hard disk, floppy disk, CD-ROMs and magnetic tapes are some examples for secondary storage.

24. What are the types of Input Devices?

There are many input devices used to feed data and instructions into the computer. Some input devices allow direct human-machine communication. Keyboard, Mouse, Light Pen, Microphone, Scanner, Magnetic Ink Character Reader (MICR), Barcode reader and Joystick are some examples for input devices.

25. What are the Applications of computers?

Computer Technology has revolutionized the business and other aspects of human life all over the world. Practically, every company, large or small, is now directly or indirectly dependent on computers for Information Processing. Computers not only save time, but also save paper work. Some of the areas where computers are being used are, Science, Education, Medicine and Health care, Engineering / Architecture / Manufacturing, Entertainment, Communication, Business Application, Publishing, Banking.

26. Define Printing Devices and its types.

The output shown on the monitor cannot be taken to other places or stored for future reference. Printers are used to get the processed information printed on paper. Printers are capable of printing alphabets, numbers, symbols and images.

Printers are classified into **Impact** printers and **Non-impact** printers depending upon the printing technique.

27. Define Operating System and its Functions.

The Operating System (OS) is integrated set of specialized programs that coordinates the various functions of the computer. The OS communicates our requests to the computer in its own language and gives you the results. In another words it acts as an interface between user and the computer hardware. The functions of OS include Device management, File management, Input-Output management, Disk management and Memory management.

28. How do you categorize the Operating System?

Modern computer OS may be categorized into four types, which are distinguished by the nature of interaction that takes place between the computer's user and his/her program during its processing. They are:

1. Single-user / Single-tasking operating system
2. Single-user / Multitasking operating system
3. Multi-user / Multitasking operating system
4. Real-time operating system

29. Define Compiler, Interpreter.

Compiler: The programs written in any high-level programming language (C or Pascal) are converted into machine language using a compiler. As a system program, the compiler translates source code (user written program) into object code (binary form).

Interpreter: An interpreter analyses and executes the source code in line-by-line manner, without looking at the entire program. In other words, an interpreter translates a statement in a program and executes the statement immediately, before translating the next source language statement.

30. List the Role of Information Technology.

Information Technology plays a vital role in the contemporary global economy. It has made a significant impact on research and development. The major areas impacted by the advent of Information Technology include IT in Business, IT in Manufacturing, IT in Mobile computing, IT in Public sector, IT in Defence services, IT in Media, IT in Education, IT in Publication.

31. What is Internet and list its important services?

The Internet is a network of many computer networks. It connects LANs, WANs and even our own personal computer. We can access any information from any place at anytime on the Internet. The Internet is often referred to as the ***'Information super highway'*****.**

Internet provides a channel for many services to be rendered online. Each service offered on the Internet is unique. Some of the most important Internet services are World Wide Web (WWW), Electronic-mail (e-mail), Electronic News groups, File Transfer Protocol (FTP), Chat services, Instant Messaging Services (IMSs), Online services, Entertainment, Bulletin boards, Peer-to-peer service.

32. Define Intranet.

An Intranet is a network within an organization. Intranet is a private computer network that uses Internet protocol technologies to securely share any part of an organization's information or operational systems within that organization.

1.1 INTRODUCTION TO COMPUTER

COMPUTER - Commonly **O**perated **M**achine **P**articularly **U**sed for **T**echnology and **E**ducational **R**esearch.

It is an electronic device which converts raw data into meaningful information.

It performs diverse operations on the information with the help of instructions to obtain desired results.

The main objective for inventing the computer was to create a **fast calculating machine**

1.1.1. Definitions

- The computer is an electronic device which converts *raw data into valid or meaningful information.*
- A device that designated in such a way, it automatically accepts and stores input data, process them and produce the desired output.
- Computer is an automatic electronic apparatus for making calculation or controlling operations that are expressible in numerical or logical terms.

i. ***Data:*** **Data** is information that has been translated into a form that is more convenient to process.

ii. ***Information:*** The processed data is called the information.

1.1.2 Basic Operations of a Computer

The computer has the following five basic operations to carry out any task

i) Input: It is the process of capturing or acquiring the information, or it is the raw data or information. By using this we can do any process.

ii) Process: It is the transformation process to convert the input in to output.

iii) Output: It is the result, which comes from the transformation process or it is the outcome of the process.

iv) Storing: It is the process of saving the data or information or instructions, so that they can be retrained and retrieved whenever they required.

v) Controlling: It is the process of directing the manner and sequence in which all the operations are to be performed.

1.2 CHARACTERISTICS TO COMPUTERS

The various characteristics of computers are as follows

1. **Speed:** Speed is the most important characteristics of computer .Computer having more speed to perform jobs instantaneously.
2. **Accuracy:** The computers are perfect, accurate and precise. Accuracy signifies the reliability of the hardware components of computers.
3. **Automatic:** A computer works automatically, once programs are stored and data are given to it, constant supervision is not required.
4. **Endurance:** A computer works continuously and will not get tired and will not suffer from lack of concentration.
5. **Versatility:** A computer can be put to work in various fields.
6. **Reduction of cost:** Though initial investment may be high, computer substantially reduces the cost of transaction.

Comparison between computer and human

S.NO	Characteristics	Computer	Human
1	Speed	Fast and excellent	Very slow
2	Accuracy	Makes no error	High possibility of errors
3	Performance	Very good	Poor
4	Instructions	Follow perfect	Chance for imperfect
5	Remember	Accurate and exact	inaccurate
6	Different Situations	Makes good	difficult
7.	Diligence	High	low

Comparison between calculator and computer

S.NO	Characteristics	Calculator	Computer
1.	Speed	fast	Much fast
2.	Performance	Simple calculation and numeric processing	Complex problem and non numeric processing
3.	Memory	Less internal memory and no permanent storage, temporary storage only	Large internal memory and large permanent storage available
4	Machine	Electronic Device	Electronic Device

1.3 HISTORY OF COMPUTERS (EVOLUTION OF COMPUTERS)

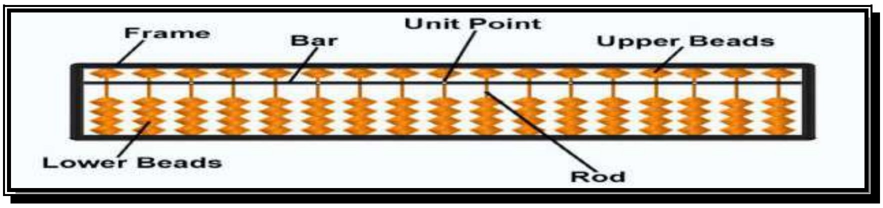
1.3.1 Abacus

In the beginning when the task was simply **counting or adding**, people used either their fingers or pebbles along lines in the sand.

In order to simplify the process of counting, people in Asia built **a counting device called abacus.**

This device allowed users to do calculations using a system of **sliding beads arranged on the rack. The beads of the counter represent digits.**

Its operation is based on the idea of the place value notation.

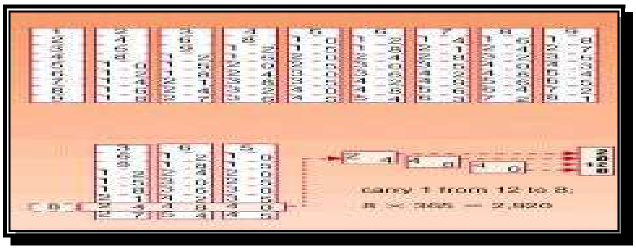


1.3.2 Napier Device

Napier invented **logarithms,** which are a technology that allows **multiplication to be performed via addition.**

The magic ingredient is the logarithm of each operand, which was originally obtained from a printed table.

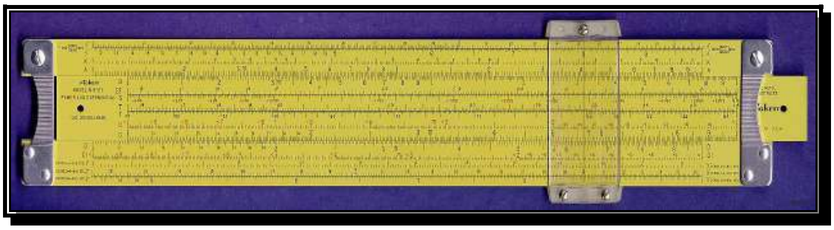
But Napier also invented an alternative to tables, where the **logarithm values were carved on ivory sticks** which are now called **Napier's bones**.



1.3.3 Slide Rule

Napier's invention led directly to the slide rule, first built in England in 1632 and still in use in the 1960's by the NASA engineers of the Mercury, Gemini, and Apollo programs which landed men on the moon.

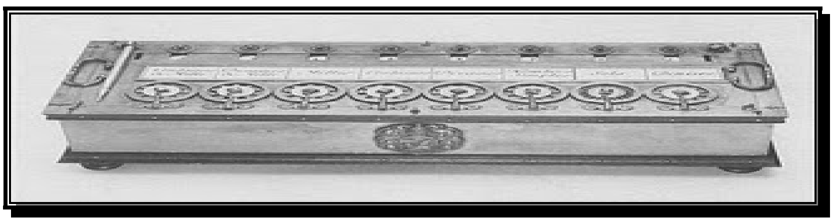
It can perform all **arithmetic and trigonometric** functions



1.3.4 Pascal Calculating Machine (Pascaline)

It was the first real desktop calculating device that could add and subtract. Pascal invented the **first functional automatic calculator.** The brass rectangular box is called a **Pascaline.**

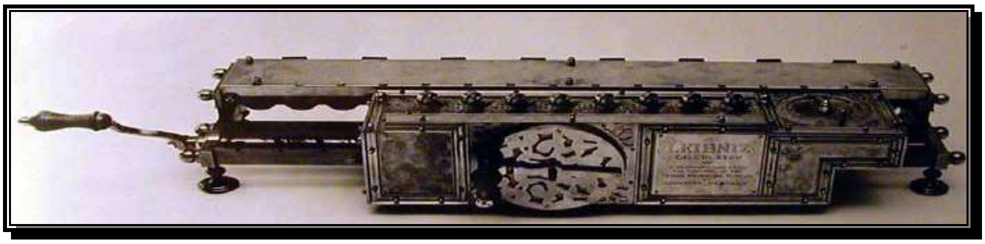
It consists of a set of toothed wheels or gears with each wheel or gear having digits 0 to 9 engraved on it. Arithmetic operation could be performed by tunings these wheels.



1.3.5 Stepped Reckoner - Leibnitz’s improved Pascal Machine

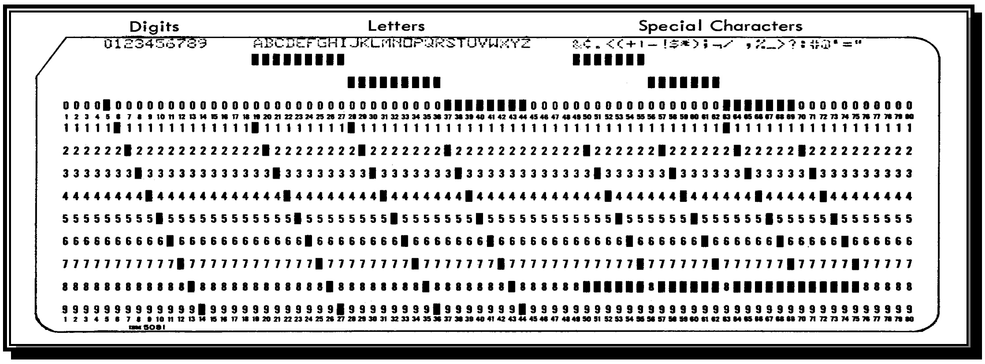
In 1694, German mathematician Gottfried wilhem von leibniz extended pascal design to perform **multiplication, division and to find square root**. This machine is known as **Stepped Reckoner**.

It was the first mass-produced calculating device, which was designed to perform multiplication by repeated addition.



1.3.6 Punched Card Machine

In 1801, a French weaver **Joseph Marie Jacquard** invented the first punched card machine. The presence and absence of the holes in the card represent the digits.



1.3.7 Charles Babbage’s Engines

In 1822 **Charles Babbage** proposed a machine to perform differential equations called a **difference Engine**.

This machine had a stored program and could perform calculations and print the results automatically.

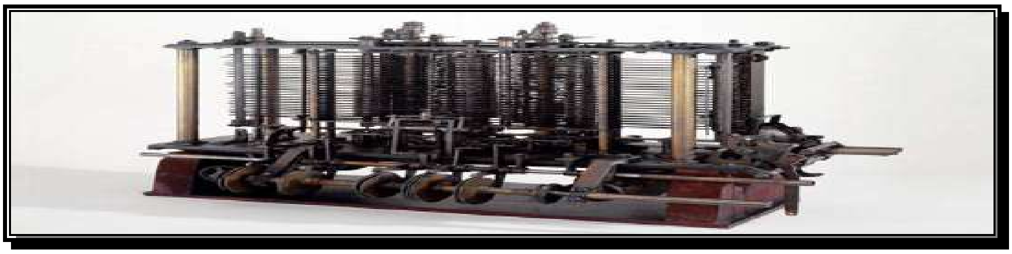
In 1833, he concentrated on **analytical engine**. The basic design of this engine included **input devices** in the form of perforated cards containing operating instructions. and a store for memory of 1000 numbers up to 50 decimal digits.

It also contains a **control unit** that allowed processing instruction in any sequence, and **output device** to produce results.

Although the analytical engine was never constructed, it outlined the basic elements of a modern computer.

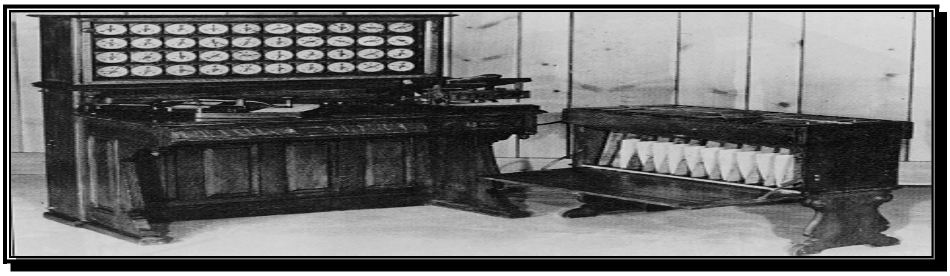
Difference Engine

Analytical Engine



1.3.8 Hollerith's Card Reading Machine

Hollerith's invention, known as the Hollerith desk, consisted of a card reader which sensed the holes in the cards, a gear driven mechanism which could count.



1.3.9 Mark-I Digital Computer

Mark I which was built as a partnership between **harvard** Aiken and **IBM** in 1944.

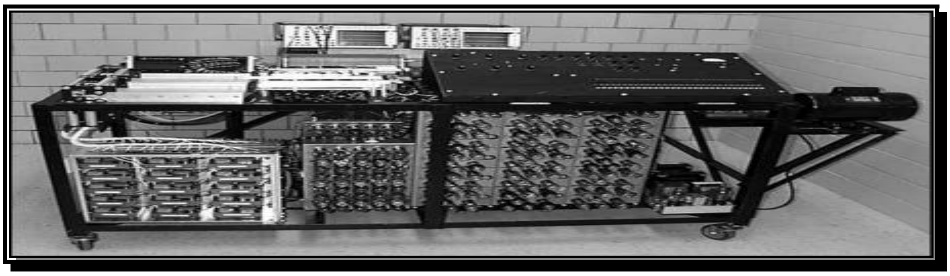
This was the **first programmable digital computer** made in the U.S. But it was not a purely electronic computer.

This electronic calculating machine used relays and electromagnetic components to replace mechanical components.

1.3.10 The First Electronic Computer (ABC)

Dr.Jhon Vincent Atanasoff and **Clifford Berry** developed the first electronic computer and called it as **ABC – A**tanasoff **– B**erry **C**omputer.

This used **Vaccum tubes** for storage, arithmetic and logical functions. This was a special purpose machine used to solve simultaneous equations.



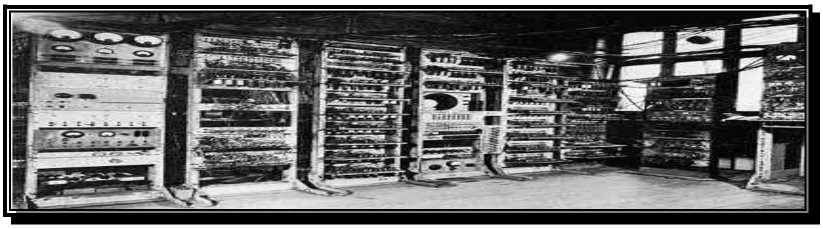
1.3.11 Electronic Numerical Integrator And Calculator (ENIAC)

ENIAC - **Electronic Numerical Integrator and calculator**. In 1946 john Eckert and John Mauchly of the Moore school of engineering at the University of Pennsylvania developed ENIAC.

It was the first electronic digital computer. ENIAC used parallel processing. This computer used **electronic vacuum tubes** to make the internal parts of the computer.

It could add or subtract 5000 times a second, a thousand times faster than any other machine.

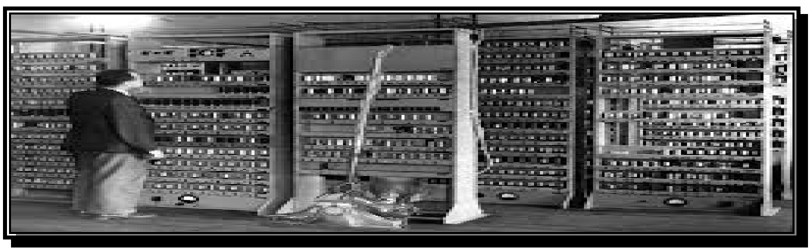
It used 18000 vacuum tubes, 70,000 resistors, 10000 capacitors and 60,000 switches and weighed 27 tones.



1.3.12 Electronic Discrete Variable Automatic Computer (EDVAC)

EDVAC - **Electronic discrete variable automatic computer.** The successor to ENIAC. It was the first stored-program computer designed by John Von Neumann in 1946.

It also had the capability of **conditional transfer of control**, that is, the computer could stop anytime and then resume operations.

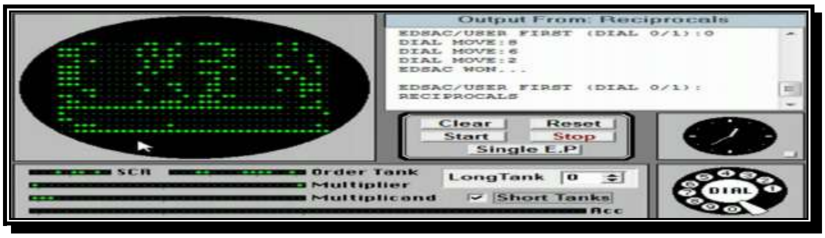


1.3.13 Electronic Delay Storage Automatic Calculator (EDSAC)

EDSAC – **Electronic Delay Storage Automatic Calculator.** In 1949 at the Cambridge University EDSAC was developed.

EDSAC was the first practical stored-program electronic computer.

This machine used **mercury delay lines** for memory and **vacuum tubes** for logic.

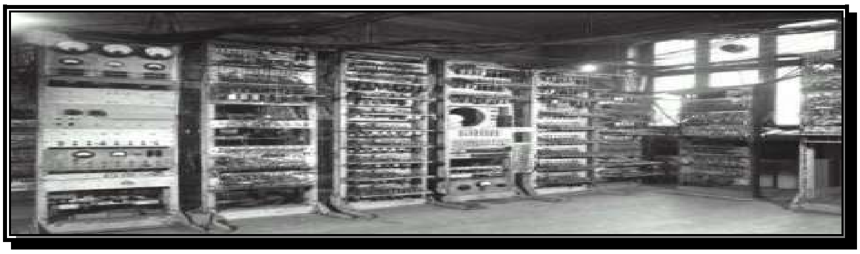


1.3.14 Manchester Mark-I

This is the small experimental computer and performs operation based on the stored program concept.

It was designed at Manchester University by a group of scientists headed by Prof. M.H.A. Newman.

It has the storage capacity of only 32 words, each of 31 binary digits. This was too limited to store data and instructions



1.3.15 Universal Automatic Computer (UNIVAC-I)

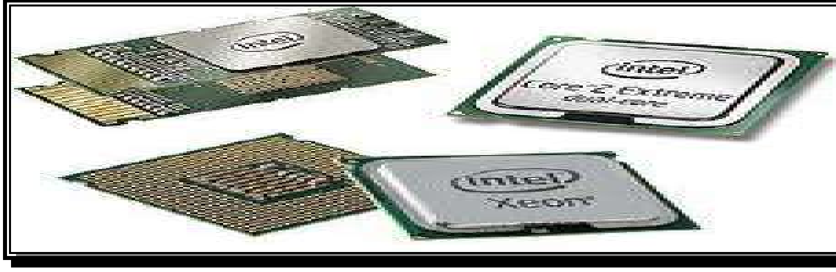
UNIVAC – **UNIVersal Automatic Computers.** The Eckert- Mauchly Corporation manufactured UNIVAC in 1951. This implementation marked the real beginning of the computer era. UNIVAC was also the first computer to **employ magnetic tape**.

1.3.16 Microprocessors

In 1970's the trend shifted towards a larger range of applications for cheaper computer systems.

A microprocessor (uP) is a computer that is fabricated on an **integrated circuit (IC).** Microprocessor a **processors on a single IC Chip**.

Computers had been around for 20 years before the first microprocessor was developed at Intel in 1971.



1.3.17 Personal Computers

The first personal computer (with Microprocessor) was developed in 1974.

In 1977, the first successful Micro Computer (PC) was developed by a young technician named **Steve Wozniak.** This was called the computer APPLE-1.

1.3.18 Power PC 600/Pentium

IMB, APPLE computers and **Motorola** co-operated in designing a Microprocessor called Power PC 600 series. Intel also designed a powerful chip in 90's called Pentium (1993).

Microprocessors such as Pentium, Power PC, Celeron, AMD, IBM, ATHLON, ZELOG, CYREX, etc, are being used as CPU of PC's since 1995.

1.4 GENERATIONS OF COMPUTERS

In computer terminology the word generation is described as a **stage of technological development or innovation.** According to technology used there are 5 generations of computers

1.4.1 First Generation of Computers (1940-56)

- First generation computer, as heavy as 7 tonnes in weight, was born with thousands of vacuum tubes.
- These computers used **vacuum tubes** for circuitry and magnetic drums for memory.
- **Input** was based on punched cards and paper tape.
- **Output** was in the form of printouts.
- They relied on **binary coded language** to perform operation operation known as **machine language.**

For example: ENIAC, EDVAC and UNIVAC.

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Vacuum Tube

Characteristics

1. Based on Vacuum Technology
2. Fast computing device of that time
3. Very large and required large space for installation.
4. Since thousands of vacuum tubes were used, they generated a large amount of heat. So air conditioning was essential
5. Non-portable and very slow equipments
6. They lacked in versatility and speed.
7. They were very expensive to operate and used a large amount of electricity.
8. These machines were unreliable and prone to frequent hardware failures. Hence constant maintenance was required.
9. Since machine language was used, they were difficult to program and use
10. Each individual component had to be assembled manually.

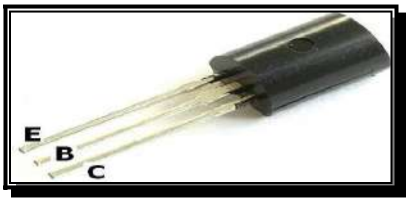
1.4.2 Second Generation of Computers (1956-63)

- Used **transistor** which were superior to vacuum tubes.
- A transistor is made up of **semiconductor material** like germanium and silicon. It usually has three leads and performs electrical functions such as voltage, current or power amplification with low power requirements.
- Since transistor is a small device, the physical size of computers was greatly reduced.

Here magnetic core was used as primary memory and magnetic disks as **secondary storage devices**.

- However they still relied on punched cards for input and printouts for output.
- **Assembly language** used mnemonics (abbreviations) for instructions rather than numbers, for example ADD for addition and MULT for multiplications.
- High level programming languages such as COBOL and FORTRAN also came into existence in this period.

For example; PDP-8, IBM 1401 and IBM 7090.



Transistor

Characteristics

1. These machines were based on transistor technology
2. Were smaller as compared to the first generation computers
3. The computational time of these computers was reduced to microseconds from milliseconds
4. These were more reliable and less prone to hardware failure. Hence such computers required less frequent maintenance.
5. were portable and generated less amount of heat
6. Assembly language was used to program computers. Hence, programming became more time- efficient.
7. Still required air conditioning.

1.4.3 Third Generation of Computers (1964-Early 1970s)

Based on **ICs**. An integrated circuit consists of a single chip with many components such as transistors and resistors fabricated on it.
ICs replaced several individual wired transistors.
This development made computers smaller in **size, reliable and efficient.**
For example: NCR 395 and B6500.

Integrated Circuit

Characteristics

1. These computers were based on IC Technology
2. They were able to reduce computational time from microseconds to nanoseconds.
3. They were easily portable and more reliable than the second generation.
4. This device consumed less power and generated less heat.
5. Air conditioning was still required
6. The size of these computers was smaller as compared to previous computers
7. Maintenance cost was quite low
8. Extensive use of high-level languages became possible

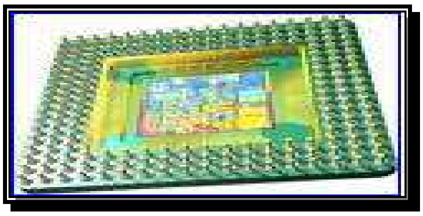
9. Manual assembling of individual components was not required, so it reduced the large requirement of people and cost.
10. Commercial production became easier and cheaper

1.4.4 Fourth Generation of Computers (Early 1970s-Till Date)

- Technology of this generation was still based on the **integrated circuit**, these have been made readily available to us because of the development of the microprocessor (Circuit containing millions of transistors)
- **A Microprocessor is built onto a single piece of silicon, known as chip.** It is about 0.5cm along one side and no more than 0.05cm thick.
- Fourth generation computers led to an era of **Large Scale Integration (LSI) and VLSI Technology.**
- LSI allowed thousands of transistors to be constructed on one small slice of silicon material whereas VLSI allows hundreds of thousands of components on to a single chip.
- ULSI (Ultra large scale integration) increased that number into millions.

During this period magnetic core memories were substituted by semi-conductor memories, which resulted in faster random access main memories

For Example: Apple II, Altair 8800, CRAY-1



Microprocessor

Characteristic

1. This generation of computers are microprocessor based systems
2. Very small
3. Cheapest among all other generations
4. Portable and quite reliable
5. These machine generate negligible amount of heat, hence they do not require air conditioning
6. Minimum maintenance is required
7. The production cost is very low
8. GUI and pointing devices enable users to learn to use the computer quickly
9. Interconnection of computers leads to better communication and resource sharing

1.4.5 Fifth Generation of Computers (1980's and Beyond)

- The process of developing fifth generation computers is **still in the development stage.**
- The speed is **extremely high** and it can perform ***parallel processing.***
- The concept of ***Artificial intelligence*** has been introduced to allow the computer to take its own decision
- However, the **expert system** concept is already in use. The expert system is defined as a computer information system that attempts to mimic the thought process and reasoning of experts in specific areas.

Characteristics

1. **Mega Chip:**
    - They use **Super Large Scale Integrated (SLSI) chips**, which will result in the production of microprocessor having millions of electronic components on a single chip.
    - In order to store instructions and information, they require a **great amount of storage capacity.**
    - Mega chip may enable the computer to approximate the memory capacity of the human mind
2. **Parallel processing:**
    - Most computers today access and execute only one instruction at a time. This is called **serial processing**.
    - However, a computer using **parallel processing** accesses **several Instructions at once** and works on them at the same time through use of multiple central processing units
3. **Artificial Intelligence**
    - It refers to a series of related technologies that tries to simulate and reproduce human behaviors, including thinking, speaking and reasoning.

1.5 BLOCK DIAGRAM OF A COMPUTER

Univ. Ques. - May 2017, May 2016, jan2016, Jan 2014, May 2013, Jan 2013 (11 marks)

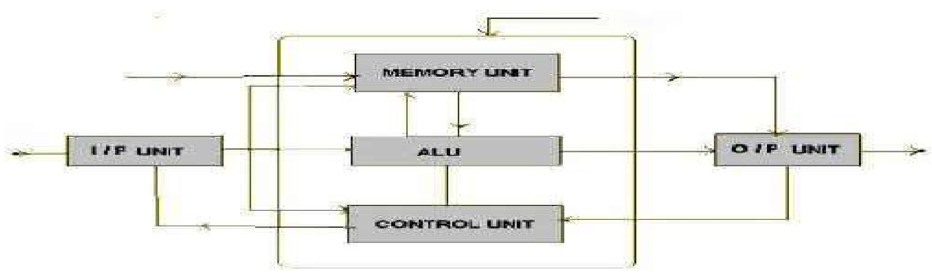
1.5.1 BASIC COMPUTER ORGANIZATION

The architecture of the computer has not changed since decades, but the technology used to accomplish those operations may vary from once computer to another computer. However, the basic computer organization remains the same for all computer systems.

The block diagram of the computer system have the following three units, each functional unit corresponds to their basic operations performed as described in details.

- Input Unit
- Output Unit
- Memory / Storage Unit
- Arithmetic Logic Unit
- Control Unit
- Central Processing Unit

These are shown in figure:



1.5.2 Input Unit

Data and instructions are entered into the memory of a computer through input devices. It **captures information and translates** it into a form that can be processed by CPU. Computer accepts input in two ways

i) Manually
ii) Directly

i) **Manually:** The user enters data into the computer by hand. E.g. Keyboard, mouse etc

ii) **Directly:** Information is fed into the computer automatically from a source document (like barcode). Some of the input devices are

INPUT DEVICES:

1. **Keyboard.**
2. **Mouse**
3. **Light Pen**
4. **Digitizer**
5. **Trackball**
6. **Joystick**
7. **OCR (Optical Character Recognizer)**
8. **MICR (Magnetic Ink Character Recognizer)**
9. **OMR (Optimal Mark Recognizer)**

1.5.3 Central Processing Unit (CPU)

- The Central Processing Unit (CPU) **takes data** and instructions from the storage unit and makes all sorts of **calculations** based on the instructions given and the type of data provided. It is then **sent back to the storage unit.**
- The CPU is the **brain** of a computer system.
- It is just like brain that takes all major decisions, makes all sorts of calculations and directs different parts of the computer functions by activating and controlling the operations.

The CPU is sub-divided into following sub-system

i) Control unit

ii) Arithmetic/logic unit (ALU) and

iii) Memory unit

- **a) Primary storage**
- **b) Secondary storage**

i) Control Unit

The control unit **controls all other units** in the computer. It directs the flow of data between Memory and Arithmetic Logic Unit. It controls and co-ordinates the entire computer system.

It fetches program instruction from the primary storage unit, interprets them and ensures correct execution of the program. It also controls the input/ output devices and directs the overall functioning of the other units of the computer.

ii) Arithmetic / Logic unit (ALU)

The task of performing operations **like arithmetic and logical operations** is called processing. ALU Comprises of two units. **Arithmetic unit** and **Logic unit**

Arithmetic Unit Contains circuitry that is responsible for performing the actual computing and carrying out the arithmetic calculations such as **addition, subtraction, multiplication and division**. It can perform these operations at a very high speed.

Logic Unit Perform **logical operation** based on the instruction. The operations are logical comparisons between data items. This unit can compare numbers, letters or special characters. Logic unit can test for 3 conditions: **equal-to condition, Less-than condition and Greater-than** condition.

iii) Memory Unit

- The process of **saving data and instructions** permanently is known as memory.
- Data has to be fed into the system before the actual processing starts. It is because the processing speed of Central Processing Unit (CPU) is so fast that the data has to be provided to CPU with the same speed.
- Therefore the data is first stored in the memory unit for faster access and processing.
- The memory unit performs the following **major functions**:
    - Data and instructions required for processing (received from input devices)
    - **Intermediate results** of processing
    - **Final results** of processing, before they are released to an output device

1. *Primary memory*

- Also known as **main memory**.
- Used to hold **running program** instructions
- Used to hold data, intermediate results, and results of ongoing processing of job(s)
- Fast in operation
- Small Capacity
- Expensive
- **Volatile** (looses data on power dissipation)
- Primary memory can be further classified into
    1. **RAM (Random Access Memory)**
    2. **ROM (Read only Memory)**

2. *Secondary memory*

- Also known as **auxiliary memory or external memory**.
- Used to hold **stored program** instructions
- Used to hold data and information of stored jobs

- Slower than primary storage
- Large Capacity
- Lot cheaper that primary storage
- **Non volatile** (Retains data even without power)

1.5.4 Output Unit

Devices used to get the response or result of a process from the computer is called output. This is the process of producing results from the input data. The outputs which can be easily understood and used by human beings are in the form of hard copy Soft copy.

Hard copy is the physical form of output is referred to as hard copy.
Soft copy is the electronic version of the output is referred as soft copy.

OUTPUT DEVICES:

1. **Monitor**
2. **Printer**
3. **Plotters**
4. **Projector**
5. **Speakers etc.,**

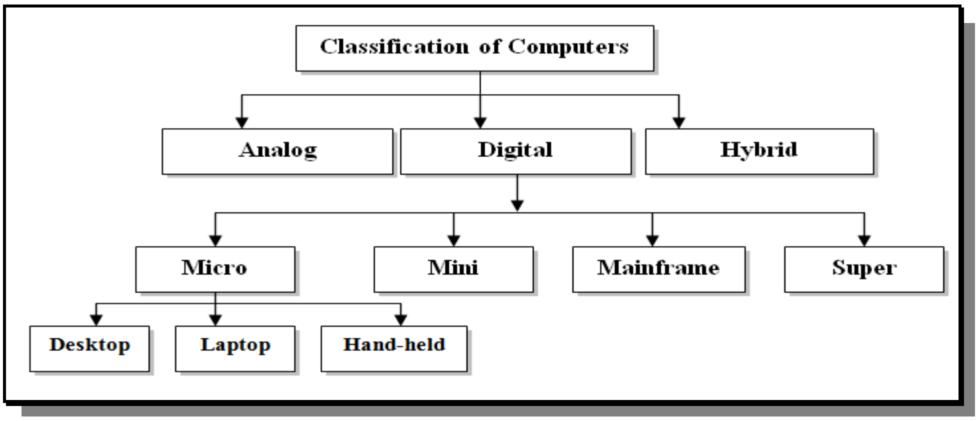
1.8 CLASSIFICATION OF COMPUTERS

Univ. Ques. May 2015, Jan 2015, May 2014, Jan 2013

Computers are available in different shape, size and weights, due to these different shapes and sizes they perform different sort of jobs from one another.

They are classified in different ways. All the computers are designed by the qualified computer architectures that design these machines as their requirements.

The following section describes how the computers are classified;



1.8.1 ANALOG COMPUTERS

- Analog is a Greek word which means similar. So in analog computers, the similarities black and white, any two quantities are measured by **electrical voltage or current**. The analog computers operate by measuring instead of counting.
- In Analog computer, the input data is continuously changing electrical or non-electrical information. Computations are carried out with physical quantities such a s Voltage, Length, Current, Temperature etc. The devices measuring such quantities are called analog devices.
- The **advantage** of analog computer is that all calculation takes place in parallel and hence it is **very fast**. It is used for engineering and scientific application. But their **accuracy is poor** as compared to digital computer.
- **Analog signals**: An analog signal is a continuous variable electromagnetic wave and it consume an infinite number of voltage or current values

1.8.2 DIGITAL COMPUTERS

- As the name implies, the digital computer with quantities represented as digits. In digital computer, both numeric and non-numeric information's are represented as strings of digits.
- In Digital Computer, the input data is discrete in nature. It is represented by **binary notation in the form of 0's and 1's**.
- Digital computers are much faster than analog computers and are more accurate. Digital computers are largely used for business and scientific applications.
- The basic operation performed by a digital computer is addition. Hence, the other operations such as multiplication, division, subtraction and exponentiation are first changed into "Addition and the compute".

Digital signal: A digital signal consists of discrete voltage levels, a light pulses, then represent either binary 1 (ON) or a binary 0 (OFF).

1.8.3 HYBRID COMPUTERS

In Hybrid Computer, an attempt is made to combine the qualities of both Analog and Digital computers. In a hybrid computer, the measuring functions are performed by the analog way while control and logic functions are digital in nature.

1.8.2 DIGITAL COMPUTERS

The digital computers are classified as

a) **Micro Computers**
b) **Mini Computers**
c) **Mainframe Computers**
d) **Super Computers**

a) Micro Computers

- **small, low cost digital computer**,
- consists of a micro processor, a storage unit, an input channel and an output channel all of which may be on **one chip** inserted into one or several PC Boards.

- In addition of a power supply and connecting cables, appropriate **peripherals** (keyboard, monitor, printer and disk drivers),
- **An operating system** and other software programs can provide a complete micro computer system

Examples: IBM-PC, Pentium 100, IBM-PC Pentium 200 and Apple macintosh

Micro computers includes

i) Desktop Computer
ii) Laptop
iii) Hand-held

i) Desktop Computers

This is the computer mostly preferred by office/home users.

- These Computers lesser in cost, small ,they are Also known as **Personal Computer**(PC)
- We can easily arrange it to fit in our home / office with all it accommodations.
- To-day this is thought to be the most popular computer in all.
- Some of the major personal computer manufacturers are ***APPLE, IBM, Del.***

ii) Laptop

- A **portable computer**, that is, a user can carry it around.
- Since the laptop computer resembles a notebook, it is also known as "**notebook**".
- Laptop computers enclosing all the basic features of a normal desktop computers.

Advantage

i) Can **use** anywhere and at anytime, especially when one is traveling.
ii) They **do not need external power** supply as a rechargeable battery is completely self contained in them

Disadvantage: Expensive compared to desktop computers.

iii) Hand-Held Computer:

A hand-held, also called **personal digital assistant** (PDA) is a computer that can conveniently be stored in a pocket (of sufficient size) and used while the user is holding it.

- They are slightly **bigger than the common calculators**.
- A PDA user generally uses a pen or electronic stylus instead of keyboard for input.
- Since these computers are easily fitted on the palm, they are also known as **palmtop computers.**
- They usually have no **disk drive, rather they use small cards to store programs and data.**
- However they can be connected to printer or a disk drive to generate output or store data.

Disadvantage: It has **limited memory** and are less powerful as compared to desktop computers

Examples: ***Apple Newton, Casio***

b) Mini Computers

In the early 1960s, **Digital Equipment Corporation (DEC)** started shipping its PDP series computer, which the press described and referred as **mini computers.**

- The mini computer is a **small digital computer whose process and storage capacity is lesser than that of a mainframe,** but more than that of micro computer.
- Its speed of processing data is in between that of a mainframe and a micro computer.
- It is about the size of a two drawer filing cabinet.
- It is used as **desktop device** that is often connected to a mainframe in order to perform the auxiliary operations.
- Minicomputer (sometimes called a **mid-range computer**) is designed to meet the computing needs for several people simultaneously in a small to medium-sized business environment.
- It is capable of supporting from **4 to about 200 simultaneous users.**
- Some of the widely used mini computers are ***PDP 11, IBM (8000 series) and VAX 7500***

c) Mainframe Computer

- Large computers both in terms of physical size as well as computations.
- They support huge number of users.
- Basically used to store and process huge amount of data
- Not all organization can offered to maintained one mainframe. Take service of Vendors
- Examples of mainframe computers Ire IBM's ESOOO, VAX 8000 and CDC 6600.

d) Super Computers

Used in scientific and engineering applications those handling huge data and do a great amount of computations

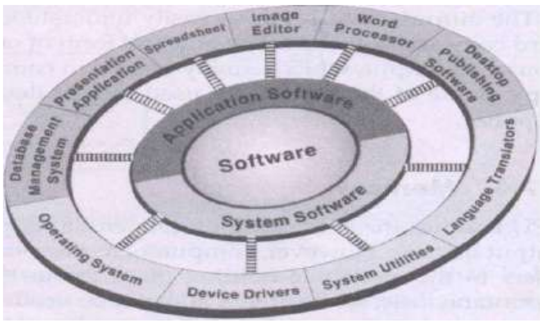
- Extremely fast in operation(@ trillion operations / seconds)
- Fast, Costliest and powerful computer available today
- Applications involves weather forecasting, military applications, electronic design etc…
- CRAY-3, Cyber 205 and PARAM are some well known super computers.

Coimarison chart for Classification of Computers:-

Type	Components	Physical Size and capacity	Cost	Usage
Microcomputers	All Components	Smallest	Cheapest	At Homes, in schools and offices
Minicomputers	Several Functional unit	Small	Cheap	In Universities, Medium sized companies, department of large companies.
Mainframe Computers	Several Separate	Large	Expensive	In large organization, universities,Government.
Super Computers	Several Separate Unit	Large	Most Expensive	In Scientific research, weather forecasting, space exploration, military defense.

1.9 CATEGORIES OF SOFTWARE

SOFTWARE: Software refers to the instructions or programs that tell the hardware what to do. It is responsible for controlling, integrating and managing the hardware components of a computer and to accomplish specific tasks.

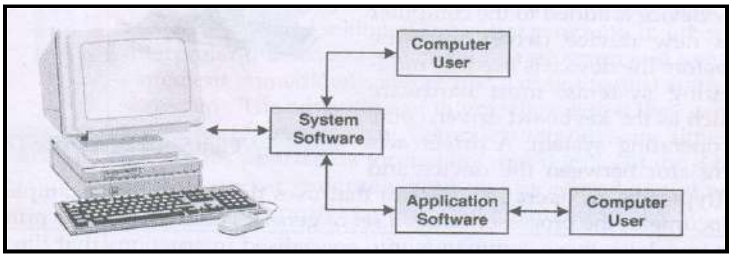


Software can be categorized in to

1. **System Software**
2. **Application Software**

1.9.1 SYSTEM SOFTWARE:

System software acts as an **interface** between the hardware of the computer and the software applications. The purpose of this software is to help the user to **run the computer** system.



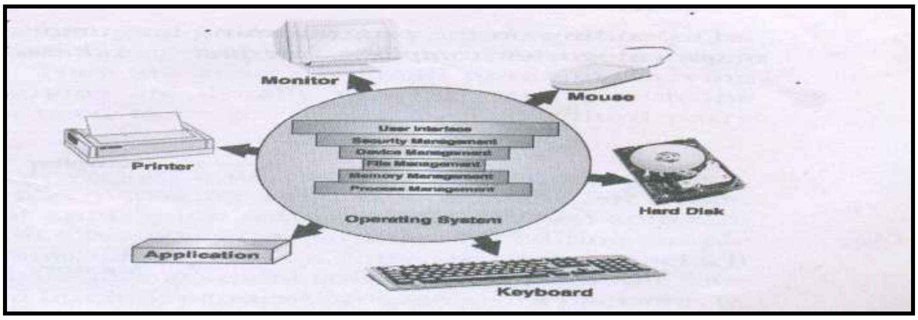
System software consists of several programs, which are directly responsible for **controlling , integrating and managing** the individual hardware components of the computer system.

- System software is more transparent and less noticed by the users, they usually **interact with the hardware** or the application and **not with the user**.
- They provide basic functions like
  1. File management
  2. Visual display
  3. Keyboard input

- *Some examples are*
  1. Operating System
  2. Device Drivers
  3. Language Translators
  4. System Utilities

a) OPERATING SYSTEM:

Operating system (OS) is an interface between hardware and user. Operating system is the first layer of software loaded into computer memory.



All other software that gets loaded for various common core service. In other words System Software provides the software platform; on top of which other programs can run (system software is the base for other software to run).

Operations of Operating System

- These **common core service** include
  1. Disk access
  2. Memory management
  3. Configuring and controlling peripheral devices
  4. Managing essential file operations, including formatting or copying disks, and renaming or deleting files
  5. Monitoring system performance (Security management)
  6. Providing a user interface

Operating system ensures that different programs executing at the same time do not interfere with each other.

Types of OS:

The Operating System is classified into various types depending upon the characteristics:

- **Single user Operating System**

In this only one user program resides in the main memory of the computer system. Once loaded it remains there till execution is complete.

- **Multi-user Operating System**

OS have been developed which can supervise the processing and manage input/output and memory allocation of more than one program simultaneously are called multi-user or concurrent OS.

- **Time-sharing Operating System**

This is another approach to handling multiple jobs. The TSOS switches the CPU from one job to another when there is a natural program break or the fixed time period has expired.

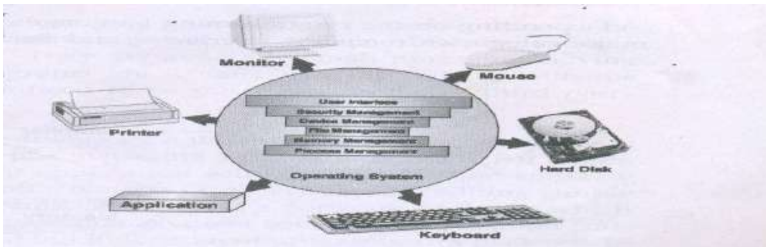


Figure 1.22 Operating System

Features of OS

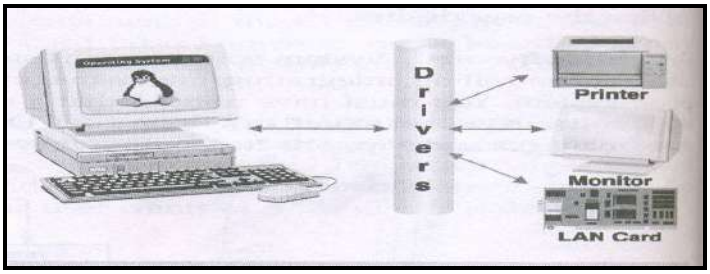
1. Process Management (program execution)
2. Memory Management
3. File management
4. Device Management
5. Security management
6. User Interface

b) DEVICE DRIVER

Device driver or **software driver** are system programs which are responsible for **proper functioning of devices**.

- Every device such as printer, monitor, mouse or keyboard, has a driver associated with it for its proper functioning.
- When ever a new device is added to the computer system, a new **device driver must be installed before the device is used**.
- In modern operating system, most hardware drivers such as keyboard **driver come with the operating system**

Ex. When a user want to print a document, the processor issues a set of generic commands to the printer driver, and the driver translate those commands into specialized instructions that the printer can understand.

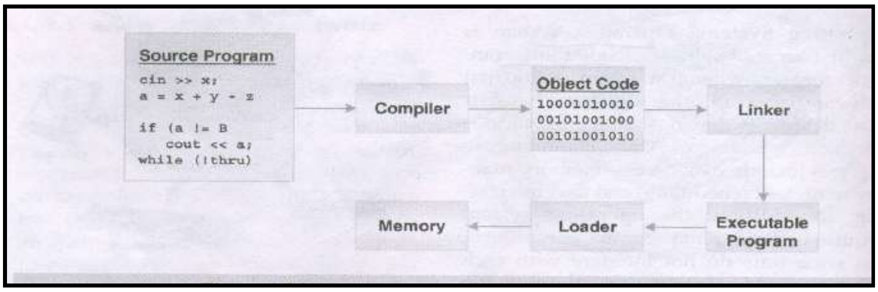


c) LANGUAGE TRANSLATOR

Computer can understand a language consisting of 0s and 1s called **machine language**

- Special programming languages called **high- level language** were developed that resemble natural languages like English.
- Therefore, **tool** was required which could translate a program written in a programming language to machine language.
- Depending on the programming language used, language translators are divided into three major categories.

1. Compiler
2. Interpreter
3. Assembler



Compiler:

*The program written in any **high level language is converted into machine language** using a compiler. Compiler translate source code (user written program) into object code (binary form)*

Interpreter:

*Interpreter **analyses and executes the source code in line-by-line manner ,** without looking at the entire program. The advantage of interpreter is that they can execute a program spontaneously. Compiler requires some time before an executable program is formed because it looks at the whole source code.*

Assembler:

*Assembly language is closest to the machine code. Assembly language is fundamentally a **symbolic representation of machine code**. The **assembly language program must be translated into machine code** by a separate program called an **assembler.***

System Utility:

- System utility program perform day to day task **related to the**
- **Maintenance** of the computer system.
- They are used to **support, enhance and secure** existing programs and data in the computer system.
- Some utility programs are usually provided along with the operating system, some are free while some need to be purchased from the 3rd party vendors.

- **Ex. For system utilitie**s:
  - ✓ Disk checkers
  - ✓ Disk cleaners
  - ✓ **Antivirus utilities**
  - ✓ Registry cleaners
  - ✓ Network managers

1.9.2 APPLICATION SOFTWARE:

- ➢ Application software is used to accomplish **specific tasks**. It consist of a single program, such as Microsoft`s Notepad(for writing and editing simple text).
- ➢ It may also consist of **a collection of programs**, often called a **software package**, which work together to accomplish a task, such as database management software.
- ➢ Application software is **dependent on system software.** Application software is **controlled by system software**, which manages hardware devices and performs back ground tasks for them.
- ➢ Application software ranges from games, calculators and word processors(text editor), paint(image Editor).

There are two types of application software:

I. General Purpose Software: - for creation of files of various types.

II. II. Computer languages: - for designing customized applications.

I .General Purpose Application Software

These are user-friendly software to help the user write letters, analyze numbers, sort files, draw pictures and even play games. It is a group of programs that provide general-purpose tools to solve specific problems.

Some of the most commonly used application software are

1. **Word processors**
2. **Spreadsheets**
3. **Image editors**
4. **Database management systems**
5. **Presentation Applications**
6. **Desktop Publishing Software**

1. Word Processors

Word processor is a software used to **compose (create), format, edit and print** electronic document. It involves not only typing, but also checking the **spelling and grammar** of the text and arranging it correctly on a page.

Example of some well known word processors are **Microsoft Word** and **WordPerfect**

2. Spreadsheets

One of the first commercial uses of computers was in **processing payroll** and other financial records. A Spreadsheet application is a rectangular grid, which **allows text, numbers and complex functions** to be entered into a matrix of thousands of individual cells.

Examples are **Microsoft Excel and Lotus 1-2-3**

3. Image Editors

Image editors are designed specifically for capturing, **creating, editing and manipulating images**. Most image editors provide a variety of special features for creating and **altering images**.

Examples are **Adobe photoshop , CorelDraw**, etc.,

4. DataBase Management System (DBMS)

Database management software is a collection of computer programs that allow **storage, modification and extraction** of information form a database in an efficient manner. It provides tools for data input, verification, storage, retrieval, query and manipulation. It also controls the **security and integrity** of the database from unauthorized access.

Examples are **Ms Access, FoxPro, Oracle etc**.,

5. Presentation Applications

Presentation graphics software allows users to create computerized slide shows that **combine text, numbers, animation, graphics, sounds, and videos**. A **slide** is an individual document that is created in presentation graphics software. A **slide show** may consist of any number of individual slides Example **Ms Powerpoint Presentation**.

6. Desktop Publishing (DTP) software

Used to describe the **creation of printed documents** using a desktop computer. It's a technique of using a personal computer to **design images** and pages and assemble type and graphics then using a laser printer or image setter to output the assembled pages onto paper, film or printing plate. This software is used for creating **magazines, books, newsletters** and so on.
Example: **Quark Express and Adobe PageMaker,** *Adobe InDesign*.

II.Computer Language Softwares:-

- Some of the popular high-level languages, that can be used to develop application softwares are, C, C++, Visual Basic, Visual C++, JAVA, and Smalltalk etc.
- The applications developed using these languages are compiled by their respective compilers and are executed to obtain desired results.

Figure 1.25 Relationship between Application and System Software

1.11 APPLICATIONS OF COMPUTERS / ROLE OF COMPUTERS

Computer is used everywhere in the world in every field of life. There are many applications of computer following list demonstrate the various Applications of computer in today's arena.

EDUCATION

Educational process leads to improve student's performance in thinking logically, formulating problem solving procedures and understanding relationship.

GOVERNANCE

IT governance derives from corporate governance and deals primarily with the connection between Business focus and IT management of an organization.

BUSINESS

The computer characteristics as speed Accuracy, Automatic, Endurance, versatility, Reduction of cost has made it an integrated part in all Business Organization. Computer used in Business
Organization for

- Payroll calculation
- Budgeting
- Managing employee database
- Maintance of stocks etc…

MILITARY

Computers are largely used in **Defense**. Some of the Military areas where a computer has been used are:

- Radar Monitoring
- Military Communication
- Missile Control
- Smart Weapons

BANKING

Banks provide following facilities:

- **Bank on-line accounting facility,** which includes
  Current Balance, Deposits, Shares, Trustee Records.

- **ATM** machine are making it even easier for customers to deals with banks.

INSURANCE:

Insurance Companies are keeping all records Up-To-Date with the Help of computers. Insurance companies are Maintaining Database of all clients' information such as

- Policy starting date
- Next due date

- Maturity date
- Interests due etc.

MARKETING

In Marketing uses of computer are following

- Advertisement
- At Home Shopping

HEALTH CARE

Computer has become important part in all medial system. The computers are being used in hospitals to keep the records of patience and medicine used in scanning and diagnosing different diseases.ECG, EEG, Ultrasounds done by Computerized Machine.

Some major Fields of Health care are:

- Diagnostic system
- Lab-diagnostic system
- Patient Monitoring System
- Pharma Information System
  (drug labels, Expiry date, etc…)

ENGINEERING DESIGN:

Computers are widely used in engineering purposes.

- CAD – Computer Aided Design
- Structural Engineering
- Industrial Engineering
- Architectural engineering

NETWORK STRUCTURES: **Computer network is an inter connected collection of computers. The computers may be in the same or different places. The connection between computers can be through copper wire, fiber optic cable , satellite etc.**

Reason for networking:-

(i)Resource sharing **:** Several computers can be connected to a single costly line resource.

(ii) Information sharing: Information on a single computer can be accessed by other computer connected in the network.

(iii) Communication: Message can be sent and received among computer connected in the network.

Network Applications:-

(i)Decision making application: Computer network helps to take decision without moving. For example a patient at one place takes treatment from a far away doctor through network.

(ii) Commercial application: Computer network help to do any type of business. By this, customer can order items through network and can pay money through credit card. This type of business is called e-business or e-commerce.

(iii)Educational applications: Computer networks help to do on-line course, to access library, to publish papers etc.

(iv) Entertainment application **:** Computer network help to hear any type of songs, play games, watch movies etc.

Benefit of network

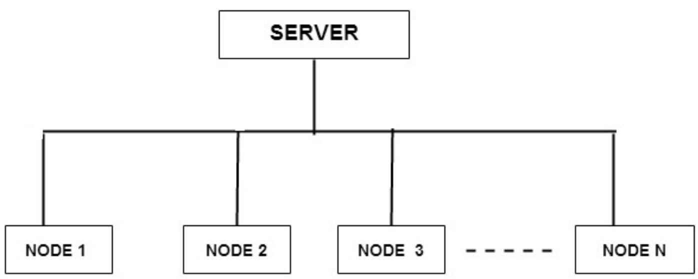
(i) It give access to remote information
(ii) It allow to share costly equipment
(iii) It allow person to person communication

Types of network

There are two important types of network. They are

(i) Local area network (LAN) (ii) Wide area network (WAN)

(i) *Local area network (LAN) :* LAN is an interconnected collection of computers placed within a small area. The connection between computer are by means of LAN cables. LAN's are usually owned and managed by a single organization. The figure given below shows a LAN connection.



Sever is a computer which contain files and program that can be shared by all nodes.

Advantages of LAN

1. All nodes share the files and programs(database) available at the server.
2. Resources such as printers , disk etc. can be shared by all nodes.
3. Data transmission cost is low.

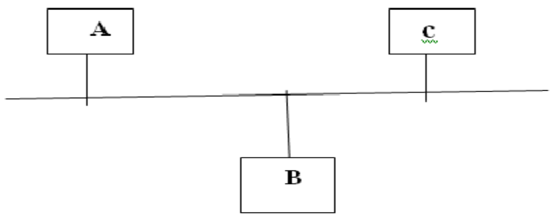
(ii) Wide area network(WAN) : WAN is an interconnected collection of computers placed at large distances. The connection between computers are by means of telephone cables or by satellite links. Using WAN we can communicate and access database in other countries.

Network topologies

Network topology is the structure of the communication network. The following are some important topologies.

(i) Bus topology

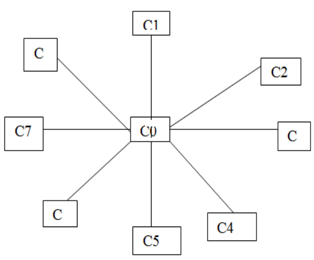
- This topology is used in LAN .
    - All the computers are connected to a single communication line. So all computer can receive the information.
- The only drawback is , if the communication line fails, the entire network is lost.



BUS TOPOLOGY

(ii) Star topology

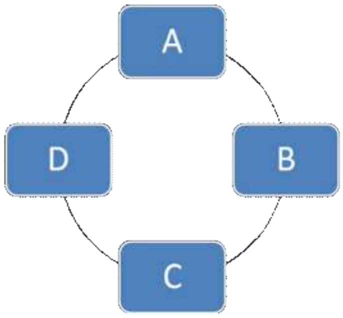
- In this one computer is at the center of the star and is called HUB.
- All other computer are attached to a hub as shown in the figure.
- Hub controls the computer attached to it.
- The only drawback is, if the hub fails the entire network is lost.



STAR TOPOLOGY

(iii) Ring topology

- Only one computer receives the signal and sends it to the next computer on the ring.
    - The only drawback is, if the communication line between two computer fails the entire network is lost.



Basics of networks

All networks require the following three elements. They are

(I)***Network services*** **:**This is provided by the computer hardware and software. Depending upon the nature of the work, we can change the input/output resources and processing power to accomplish their goal.

(ii) ***Transmission media :*** This is the pathway for connecting one computer with the other. Connection can be made by physical wires or by wireless techniques.

(ii)***Protocols*** **:** Protocols are rules or standard available for computer to computer communication. The widely used protocol is TCP/IP.

Communication network services

The following are some important network services. They are

- ✓ ***File service :*** This service improves the efficient storage and retrieved of network of network data.
- ✓ ***Print services:***This service controls and manages the access to printers and fax instruments connected to the networks.
- ✓ ***Message services:***This service helps to store, access and deliver text, binary, graphic, digitized video and audio data.
- ✓ ***Application services:***This service runs software for network users.

Forms of data transmission

There are two forms of data transmission. They are

(i) ***Analog data transmission :*** In this type transmission, the data are transmitted in a continuous wave form. The telephone communication is a analog transmission.

(ii) ***Digital transmission :*** In this type transmission, the data are transmitted in a distinct electrical state of on and off. These states are represented by 1 and 0.

1.13 INTERNET

INTERNET: Internet is defined as the network of network. Internet is a network which covers the whole world. To this millions of computers are connected.

Getting Connected To connect the following with the internet, the following hardware components are required.

1. A computer
2. Telephone connection
3. Modem
4. Internet account from any internet service provider (ISP)

Components of Internet Account Internet account consists of

a. Username - A name to identify the user

b. Password - To protect unauthorized usage of internet.

c. IP address - The address provided by the service provider.

Browsing Searching for the Information on the internet is called browsing. To browse, we need a special program called web browser.

Commonly used web browsers are:

- Internet Explorer
- Netscape Navigator
- Alta Vista
- Google Chrome etc…

Uses of the internet

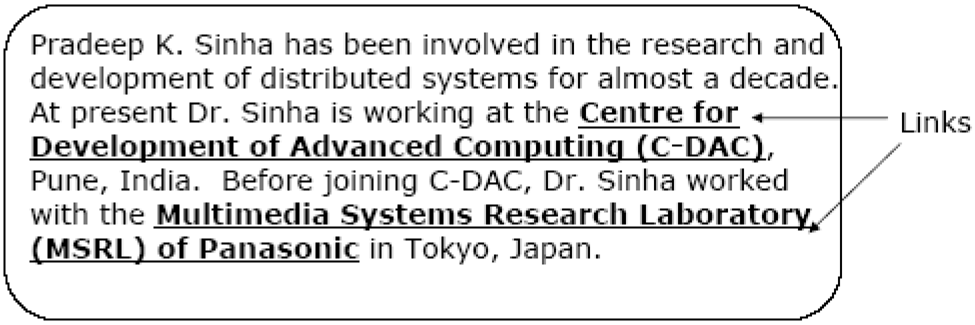
Some important current strategic uses of the Internet are:

- On-line communication
- Software sharing
- Exchange of views on topics of common interest
- Posting of information of general interest
- Product promotion
- Feedback about products
- Customer support service
- On-line journals and magazines
- On-line shopping
- World-wide video conferencing

1.13.1 BASIC INTERNET TERMINOLOGY: Univ. Ques. May 2016

Webpage:

- The World Wide Web consists of files, called pages or **web pages**, which contain information and links to resources throughout the internet.
- A web page is an **electronic document** written in a computer language called **HTML** (Hypertext Markup Language).
- web page is also known as **HTML page**
- Web pages are linked together through a system of connections called **hyperlink**, which enable the use to jump from one page to another.



Website:

- Website is a set of related **WebPages**, images, videos or audios, published by an organization or individual.
- Website is accessed by its own address (also known as **URL** or Uniform resource Locator)

Home page:

- The homepage is the URL or local file that automatically loads when a web browser starts and when the browser's "home" button is pressed.
- Home page is also known as **index page.**

Web browser:

A software program that allows users to access the Internet
Eg: **Microsoft Internet Explorer and Netscape Navigator**

Browsers are of two types:

1. Graphical a user interface for computers which enables people to see color, graphics, and hear sound and see video, available on Internet sites. These features are usually designated by underlined text, a change of color, or other distinguishing feature; sometimes the link is not obvious, for example, a picture with no designated characteristic. Examples are **Netscape** and **Internet Explorer**

2. Non-graphical a user interface for computers which allows you to **read plain text,** not pictures, sound, or video, on the Internet. It is strictly text based, non-Windows, and does not place high memory demands on your computer. An example is **lynx** (http://lynx.browser.org/)

URL:

- Each web page has a unique address called a uniform Resource Locator(URL) that identifies its location on the internet.
- Format of URL: (consist of **six parts**)

Protocol:// <World Wide Web > <Domain Name> <Domain Type>/path/ File Name

Eg: http://www.Google.com/fop/notes/unit1.htm

Protocol: http
Domain Name: Google
Domain Type: com
Path: fop/notes/

File Name: unit1.htm

Domain Type	Description
com	Commercial and profit organisation
Edu	Educational purpose
gov	Government agencies
mil	Military sites
net	Internet infrastructure and service providers
org	Non-profit organization
uk	Country code – united kingdom
de	Germany
au	Australia
in	India

- First part of address, the part before the colon, is the access methods.
- Most of the time http(hyper Text transfer protocol) is used

protocol	Description
http	Hyper text transfer protocol
ftp	File transfer protocol
usenet	Network news transfer protocol()
mailto	Mail Server
telnet	Telnet for accessing remote computers

Hypertext – text that is non-sequential, produced by writing in HTML (**H**yper**t**ext **M**arkup **L**anguage) language. This HTML coding allows the information (text, graphics, sound, video) to be accessed using HTTP (Hypertext Transfer Protocol).

Internet Service Provider: A company, which provides users with an access to the Internet, is known as an Internet service provider or Internet access provider. ISP provides the user software package, username, password and access phone number. Equipped with a modem, the user can then log on to the Internet and browse the web.

While choosing the ISP **some factors** should be considered.

1. The user should check the speed and consistency of the internet access.
2. user should check ISP's reliability (consistent access to the internet or does it have frequent disconnections)
3. Price of the ISP should be also considered.

Web server: A web server is a computer program that accepts HTTP requests from web clients and provides them with HTTP responses.

Download and upload: Download refers to the activity of moving or copying a document program from internet or other interconnected computers to ones own computer. It is the process of **pulling information**. For example while accessing a website, the HTML code and graphics must be downloaded from a remote server onto the users' computer.

Upload is just opposite of download. Here the user moves or copies a document, program or other data from his/ her computer to the internet. For example a software company may upload a demonstration of its new software onto the web.

Online and offline:

- The term online is commonly referred to as connected to the WWW via internet.
- Offline, it refers to the actions performed when the user is not connected, via telecommunications to another computer or a network like the internet

Blog - A blog is information that is instantly published to a Web site. Blog scripting allows someone to **automatically post information** to a Web site. The information first goes to a blogger Web site. Then the information is automatically inserted into a template tailored for your Web site.

Bookmark – a way of storing your favorite sites on the Internet. Browsers like Netscape or Internet Explorer let you to categorize your bookmarks into folders.

CGI (**C**ommon **G**ateway **I**nterface script) – a specification for transferring information between a Web server and a CGI program, designed to receive and return data

Chat – real-time, synchronous, text-based communication via computer.

Cookie – Information (in these case URLs created by a Web server and stored on a user's computer. People can set up their browsers to accept or not accept cookies.

FTP – Using file transfer protocol software to receive from upload) or send to (download) files (text, pictures, spreadsheets, etc.) from one computer/server to another.

Home page - Generally the first page retrieved when accessing a Web site. Usually a "home" page acts as the starting point for a user to access information on the site. The "home" page usually has some type of table of contents for the rest of the site information or other materials. When creating Web pages, the "home" page has the filename "index.html," which is the default name. The "index" page automatically opens up as the "home" page.

HTML – A type of text code in Hypertext Markup Language which, when embedded in a document, allows that document to be read and distributed across the Internet.

HTTP – The hypertext transfer protocol (http) that enables html documents to be read on the Internet.

Hyperlink – Text, images, graphics that, when clicked with a mouse (or activated by keystrokes) will connect the user to a new Web site. The link is usually obvious, such as underlined text or a "button" of some type, but not always.

Instant Messaging (IM) – a text-based computer conference over the Internet between two or more people who must be online at the same time.

IP Address – (Internet Protocol) The number or name of the computer from which you send and receive information on the Internet.

Modem – A device that connects your computer to the Internet. The modem translates computer signals to analog signals which are sent via phone lines. The telephone "speaks" to the computer/server which provides your Internet access.

GETTING CONNECTED TO INTERNET APPLICATIONS:

The basic requirement for getting online are
1. A TCP/IP enabled computer with a web browser
2. An account with an ISP
3. A telephone line plugged to a suitable socket
4. A modem to connect the computer to the telephone line.

Computer

- Computer must have at least 386-microprocessor chip with a minimum of 16MB of RAM.
- For good browsing experience, use a faster chip (Pentium III / Pentium IV) with more RAM (128/256 MB).
- In addition a color monitor, with atleast 640 X 480 resolution and a capability of displaying a minimum of 256 colors.
- The system should also possess a hard disk, with at least 200 MB of free space, to store internet software and temporary internet file.
- Since the web is a multimedia medium, you can also enjoy sound on your system with a sound card and a pair of speakers.

Modem (Modulator - demodulator):
- A computer system must have a device called modem, which enables a computer to transmit data over telephone lines.
- A modem is a hardware, which converts digital data into analog signals (that is, modulation) that can be sent over an analog telephone line and convert the analog signal back into digital data (that is, demodulation)
- Modem speeds are measured in kilobits per sec (Kbps). Modern modem supports 28Kbps to 56Kbps
- Modems are of two types :
    1. Internal modem
    2. External Modem
- A modem that sits inside the computer is called an *internal* modem, while modems that sit exterior, are *external* modems.

1.13.2 INTERNET SERVICES: / APPLICATIONS OF INTERNET:

Univ. Ques. May 2017, Nov 2016, Jan 2016, May 2015, Jan 2015, May 2013, Jan 2013

1. World Wide Web

A system of Internet servers that support specially formatted documents. The documents are formatted in a markup language called HTML (**HyperText Markup Language**) that supports links to other documents, as well as graphics, audio, and video files. This means you can jump from one document to another simply by clicking on hot spots. Not all Internet servers are part of the World Wide Web.

There are several applications called Web browsers that make it easy to access the World Wide Web; Two of the most popular being Firefox and Microsoft's Internet Explorer.

2. e-mail

Electronic mail ***is*** the transmission of messages over communications networks. The messages can be notes entered from the keyboard or electronic files stored on disk. Most mainframes, minicomputers, and computer networks have an e-mail system. Some electronic-mail systems are confined to a single computer system or network, but others have gateways to other computer systems, enabling users to send electronic mail anywhere in the world. Companies that are fully computerized make extensive use of e-mail because it is fast, flexible, and reliable.

Sent messages are stored in electronic mailboxes until the recipient fetches them. All online services and Internet Service Providers (ISPs) offer e-mail, and most also support gateways so that you can **exchange mail with users of other systems.**

3. File transfer Protocol - FTP

File *T*ransfer *P*rotocol, the protocol for exchanging files over the Internet. FTP works in the same way as HTTP for transferring Web pages from a server to a user's browser and SMTP for transferring electronic mail across the Internet in that, like these technologies, FTP uses the Internet's TCP/IP protocols to enable data transfer. FTP is most commonly used to download a file from a server using the Internet or to upload a file to a server (e.g., uploading a Web page file to a server).

4. Telnet

A terminal emulation program for TCP/IP networks such as the Internet. The Telnet program runs on your computer and connects your PC to a server on the network. You can then enter commands through the Telnet program and they will be executed as if you were entering them directly on the server console. This enables you to control the server and communicate with other servers on the network. To start a Telnet session, you must log in to a server by entering a valid username and password. Telnet is a common way to remotely control Web servers.

5. Internet Relay Chat -IRC

***I*nternet *R*elay *C*hat,** a **chat system** developed by Jarkko Oikarinen in Finland in the late 1980s. IRC has become very popular as more people get connected to the Internet because it enables people connected anywhere on the Internet to join in live discussions. Unlike older chat systems, IRC is not limited to just two participants.

To join an IRC discussion, you need an **IRC** *client* and Internet access. The IRC client is a program that runs on your computer and sends and receives messages to and from an IRC **server**.

The IRC server, in turn, is responsible for making sure that all messages are broadcast to everyone participating in a discussion. There can be many discussions going on at once; each one is assigned a unique **channel**.

6. Usenet

USENET is a worldwide bulletin board system that can be accessed through the Internet or through many online services. The USENET contains more than 14,000 forums, called **newsgroups**, which cover every imaginable interest group. It is used daily by millions of people around the world.

7. Chatting and Instant Messaging:

Chat programs allow users on the internet to communicate with each other by typing in real-time. Note that even though chatting is based on standardized IRC system, unlike the user does not need to have any special software to connect to any chat rooms.

A variation of chat is instant messaging where a user on the web can contact another currently logged in and type a conversation. To avail this Internet service, the user must have software called instant messenger installed on the system.

8. Internet Telephony:

Internet telephony is the use of Internet rather than traditional telephone company infrastructure, to exchange spoken or other telephonic information. It consists of hardware and software that enable people to use the Internet as a transmission medium phone call. There are many Internet telephony applications available. Some, such Cool Talk and NetMeeting come bundled with popular web browsers.

9. Video Conferencing:

Video conferencing uses the same technology as IRC, but also provides sound and video pictures. It enables direct face-to-face communication across network.

10. Commerce through Internet:

Today business is taking place through electronic telecommunication media. Nowadays, various organizations offer the facility of shopping online. This type of business model is known as Electronic Commerce or E-commerce. E-commerce refers to buying and selling goods and online.

When discussing about e-commerce, we normally talk about virtual shopping malls, up stores where the user virtually visits and selects the product(s) of his/her choice. After selecting the desired product(s), the payment for the purchase is done. Online payment can be through credit card. Some of the well-known e-commerce sites are www.amazon.com and www.ebay.com.

Mobile commerce or M-commerce refers to transactions through mobile phone network and data connection that result in the transfer of value in exchange for goods and services.

1.14 INTRANET:

Intranet is a network connection within a company or organization using tcp/ip and other internet standard protocols

APPLICATIONS OF INTRANET:

1. Document publication applications:

The first application that always comes to mind for intranets in and of themselves is the publication and distribution of documents. This application allows for paperless publication of any business information that is needed for internal employees or external customers or suppliers. Any type of document may be published on an intranet: policy and procedure manuals, employee benefits, software user guides, online help, training manuals, vacancy announcements-the list goes on to include any company documentation.

2. Electronic resources applications:

In the past it has not been easy to share electronic resources across network nodes. Employees have had problems sharing information for various reasons including software version inaccuracies and incompatibilities. Intranets provide the means to catalog resources online for easy deployment across the network to any authorized user with the click of a mouse. Software applications, templates, and tools are easily downloaded to any machine on the network.

3. Interactive communication applications:

Two-way communications and collaboration on projects, papers, and topics of interest become easy across the intranet. Types of communications that are enhanced and facilitated include e-mail, group document review, and use of groupware for developing new products.

4. Support for Internet applications:

Even though organizational full-service intranets are the next step in enterprise-wide computing and have enough value to make them desirable simply for the organizational benefits they bring, they are also necessary for supporting any Internet applications that are built.

1.15 STUDY OF WORD PROCESSOR:

Word processor (MS Word):

- MS Word is application software that can be used to **create, edit, save and print** personal as well as professional documents in a very simple an efficient manner.
- Saved with the extension **.doc**
- MS Word is an important tool that is mainly designed for word processing.
- MS Word is not the only word processing program available in the market.
- There are many other word processing applications available such as **Open Office Writer** and **Google Docs**

ACCESSING MS WORD:

1. Using Menu:

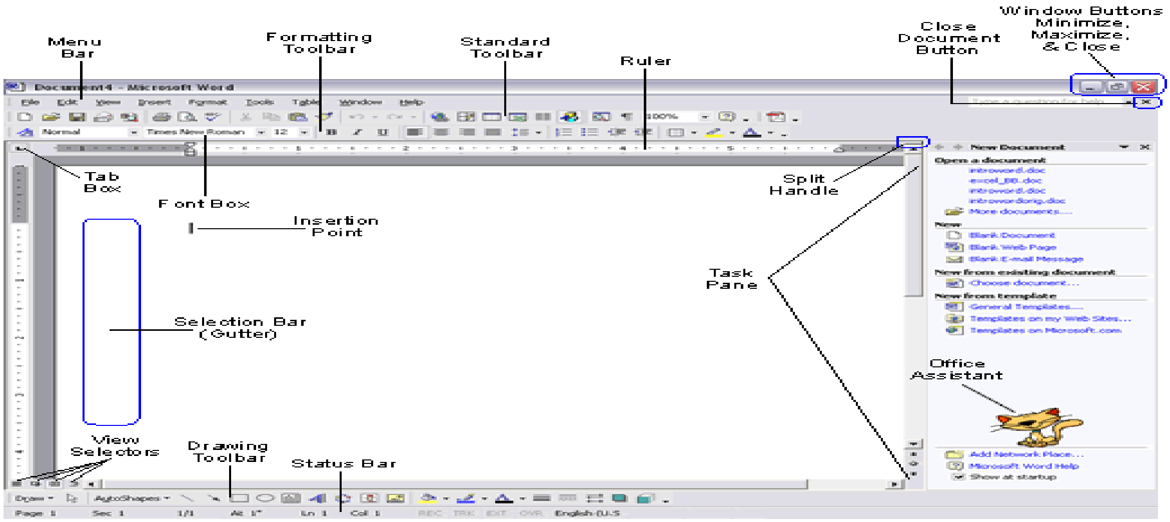
Start□ Programs□Microsoft Office□ Microsoft Office Word 2003

2. Run command

Start□ Run to display the Run dialog box

Type **winword** in the open text box and click OK

Components of the Word Window



Component	Functionality or Purpose of the Component
Menu Bar	Contains File, Edit, View, Insert, Format, Tools, Table, Window and Help menus
Title Bar	It display the name of the currently opened MS word.
Standard Toolbar	Contains icons for shortcuts to menu commands.
Formatting Tool Bar	Contains pop-up menus for style, font, and font size; icons for boldface, italic, and underline; alignment icons; number and bullet list icons; indention icons, the border icon, highlight, and font color icons.

Ruler	Ruler on which you can set tabs, paragraph alignment, and other formats.
Insertion Point	Blinking vertical bar that indicates where text you type will be inserted. Don't confuse the insertion point with the mouse I-beam. To move the insertion point, just click the mouse where you want the point moved.
End-of-File Marker	Non-printing symbol that marks the end of the file. You cannot insert text after this mark.
Selection Bar (Gutter)	Invisible narrow strip along the left edge of the window. Your mouse pointer changes to a right-pointing arrow when it is in this area. It is used to select a line, a paragraph, or the entire document.
Split Handle	Double-click to split the window in two (to view different portions of the same file). Double-click to return to one window
Status Bar	Displays page number, section number, and total number of pages, pointer position on page and time of day.
Task Pane	Displays and groups commonly used features for convenience
Office Assistant	An animated character that can provide help and suggestions. There are multiple characters to choose from, and it is possible to turn the Office Assistant off.

Advantages of Using a Word Processor

- It won't improve your typing speed
- Does make it easier to correct and modify the document
- Word wrap eliminates concern about where to end a line
- Particularly powerful for editing a document
- Easy insertion and deletion of text
- Ability to combine text from two or more documents
- Additional support with spelling and grammar checkers
- Easy to modify to the output formatting

Word Processing Terminology

- Bold - produces dark or heavy print
- Center - centering text evenly between the margins
- Edit - change or modify the document
- Font - type style and size
- Format - defines how printed document will appear; e.g. underline, boldface, print size, margin settings, line spacing, etc.
- Grammar checker - checks for grammar, phrasing, capitalization
- Justified - text is evenly aligned on both left and right margin
- Spelling checker - checks words or entire document for spelling

- Text data - any number, letter or symbol you can type on the keyboard
- Word wrap - automatic adjustment of the number of characters or words on a line; eliminates need to press Enter key

Loading MS-Word

- Start, Programs, Microsoft Word or
- Double-click on Word shortcut on the desktop
- From the Microsoft Office Shortcut Bar, click on New Office Document, then Blank Document

Creating a New Document

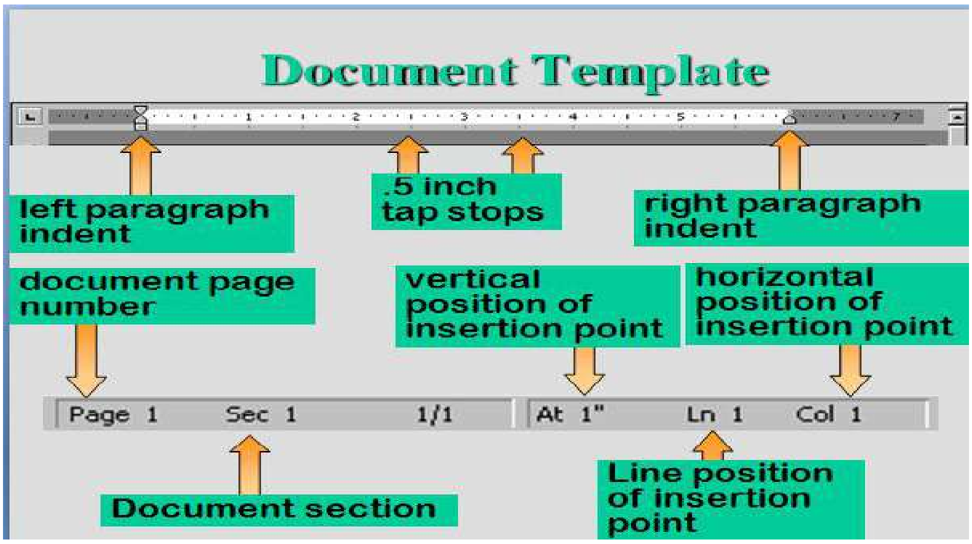
- Planning - understand document's purpose and what it will say
- Entering - entering contents by typing into word processor
- Editing - making changes to document, to correct errors and enhance content
- Formatting - enhancing to make more readable and attractive, such as boldface text, italics and bullets
- Previewing and Printing - displaying on screen how document will look printed, allows for final changes before printing

Entering and Editing Text

- New Word document like a blank piece of paper, but with many predefined or *default* settings
- Default settings contained in the document template

Document Template

- Every Word document based on a document template
- Settings are stored in the Blank document template
- Other templates can be used to create different styles of memos, letters and reports



AutoCorrect:

- Part of MS-Office *IntelliSense* feature
- Makes basic assumptions about text user is typing

- Uses assumptions to automatically identify and correct an entry
- Corrects for capitalization and spelling
- Can prompt user by finishing common phrases

Spelling and Grammar Checkers

- Automatically advises user of misspelled words or incorrect grammar
- Spelling Checker compares typed words to those in a main dictionary
- Those failing to match are identified as misspelled
- User can then correct herself or choose from menu of suggested words
- Grammar Checker looks for incorrect verb and tense agreements, verb forms, commonly confused words and others
- Words and phrases identified with a wavy green line
- User can then edit, or choose from a menu of suggested changes

Inserting and Deleting Blank Lines

- If insertion point is on a line above another line, and at the beginning or end of the line, press Enter key to
- Two ways of deleting a blank line
  - ✓ Pressing the delete key
  - ✓ Click on Edit, Clear

Inserting Characters

- Move your cursor to the insert area
- Type in the text desired
- Add spaces if necessary

Deleting Words : Position your insertion cursor and press the Control Delete key

Word Wrap:

- Feature that automatically decides where to end a line
- Text is "wrapped" to next line based on margin settings
- Saves time - pressing Enter key only when starting new paragraph or inserting blank line

Opening, Closing and Saving Files

- Closing
  - File, Close menu command
  - File Close button
- Saving
  - File, Save and Save menu commands
  - Diskette file save icon

Moving Through a Document

- Moving mouse
- PgUp and PgDn keys
- Vertical scroll bar and buttons
- Horizontal scroll bar and buttons

Editing Key Summary

WORD SHORTCUT KEYS

See our Microsoft Word page for additional help and information.

Shortcut Keys	Description
Ctrl + A	Select all contents of the page.
Ctrl + B	Bold highlighted selection.
Ctrl + C	Copy selected text.
Ctrl + X	Cut selected text.

Ctrl + Shift + F	Change the font.
using thumbnail images	Increase selected font +1.
Ctrl +]	Increase selected font +1.
Ctrl + Shift + <	Decrease selected font -1.
Ctrl + [	Decrease selected font -1.
Ctrl + Shift + *	View or hide non printing characters.
Ctrl + <left arrow>	Moves one word to the left.
Ctrl + <right arrow>	Moves one word to the right.
Ctrl + <up arrow>	Moves to the beginning of the line or paragraph.

Ctrl + Del	Deletes word to right of cursor.
Ctrl + Backspace	Deletes word to left of cursor.
Ctrl + End	Moves the cursor to the end of the document.
Ctrl + Home	Moves the cursor to the beginning of the document.
Ctrl + Spacebar	Reset highlighted text to the default font.
Ctrl + 1	Single-space lines.
Ctrl + 2	Double-space lines.
Ctrl + 5	1.5-line spacing.
Ctrl + Alt + 1	Changes text to heading 1.
Ctrl + Alt + 2	Changes text to heading 2.
Ctrl + Alt + 3	Changes text to heading 3.
Ctrl + F1	Open the Task Pane.
F1	Open Help.
Shift + F3	Change the case of the selected text.
Shift + Insert	Paste.
F4	Repeat the last action performed (Word 2000+)
F5	Open go to window.
F7	Spell check selected text and/or document.

Shift + F7	Activate the thesaurus.
F12	Save as.
Shift + F12	Save.
Alt + Shift + D	Insert the current date.
Alt + Shift + T	Insert the current time.

Mouse Shortcuts	Description
Click, hold, and drag	Selects text from where you click and hold to the point you drag and let go.
Double-click	If double-click a word, selects the complete word.
Double-click	Double-clicking on the left, center, or right of a blank line will make the alignment of right aligned.
Double-click	Double-clicking anywhere after text on a line will set a tab stop.
Triple-click	Selects the line or paragraph of the text the mouse triple-clicked.
Ctrl + Mouse wheel	Zooms in and out of document.

Selecting Text

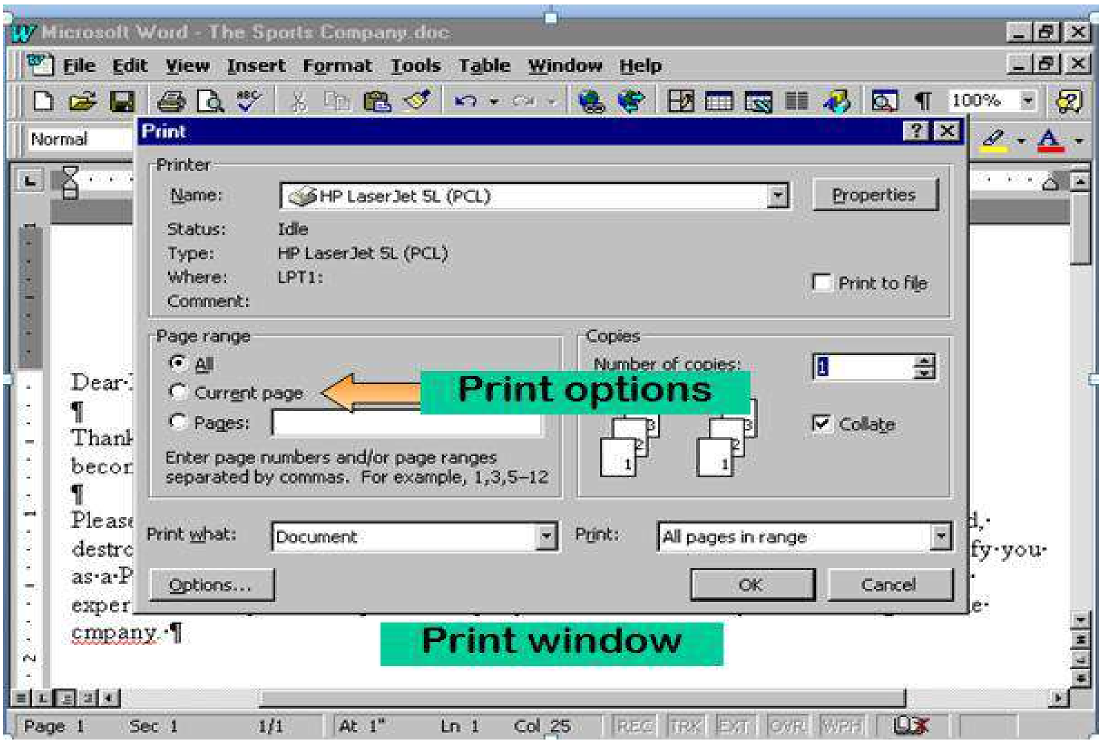
- Word- Double-click in the word
- Sentence- Press Ctrl and click within the sentence
- Line- Click in the selection bar next to the line
- Multiple lines-Drag in the selection bar next to the lines
- Paragraph- Triple-click on the paragraph or double-click in the selection bar next to the paragraph

Undoing Editing Changes

- Delete the change
- Click on the Undo button
- Click on the Edit, Undo menu

Printing a Document

1. View onscreen how document will look printed out
2. Each page displayed in a reduced sized so that you can see the layout

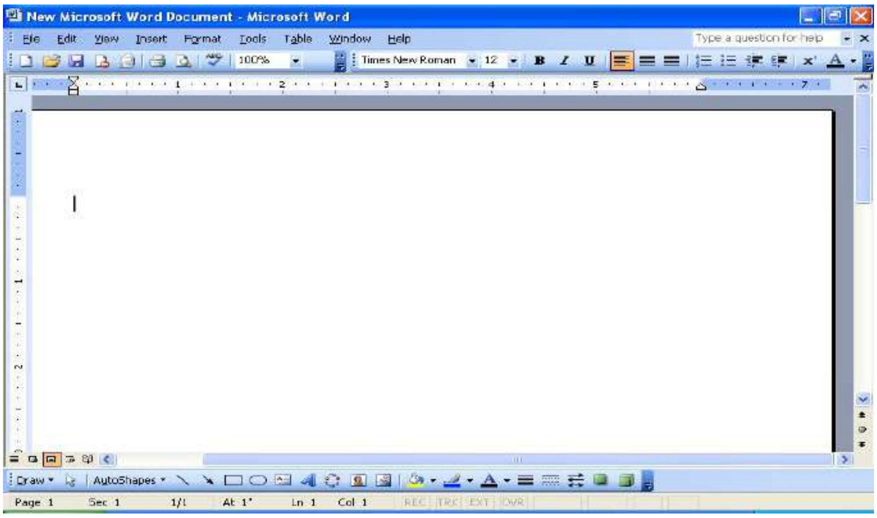


BASIC OPERATIONS PERFORMED IN MS WORD:

- Creating a document
- Saving a document
- Editing a document
- Formatting a document
- Printing a document

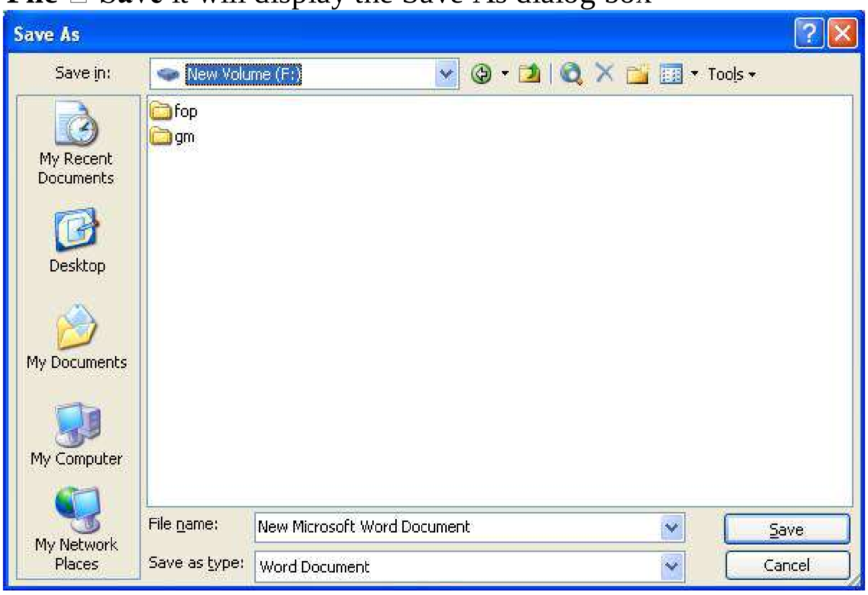
Creating a Document

- Open MS Word
- The document will have a vertical blinking bar, called as insertion point or Cursor point. Insertion point usually defines the place in the document window from where we start typing the text.



Saving a Document:

- **File** □ **Save** it will display the Save As dialog box



- Select the location and save the document.

Save	Save As
When you save a new document for the first time, Word displays a dialog box (see figure, below). Select where you want to save your document and give it a name. When you save an existing document that you have been editing, the newly saved version is written over the older version.	This command always displays a dialog box where you can choose a document name and disk (see figure, below). Use the Save As. command wheneve you want to save a copy of the current document under a different name or in a different folder (or disk). The newly saved copy becomes the active document.

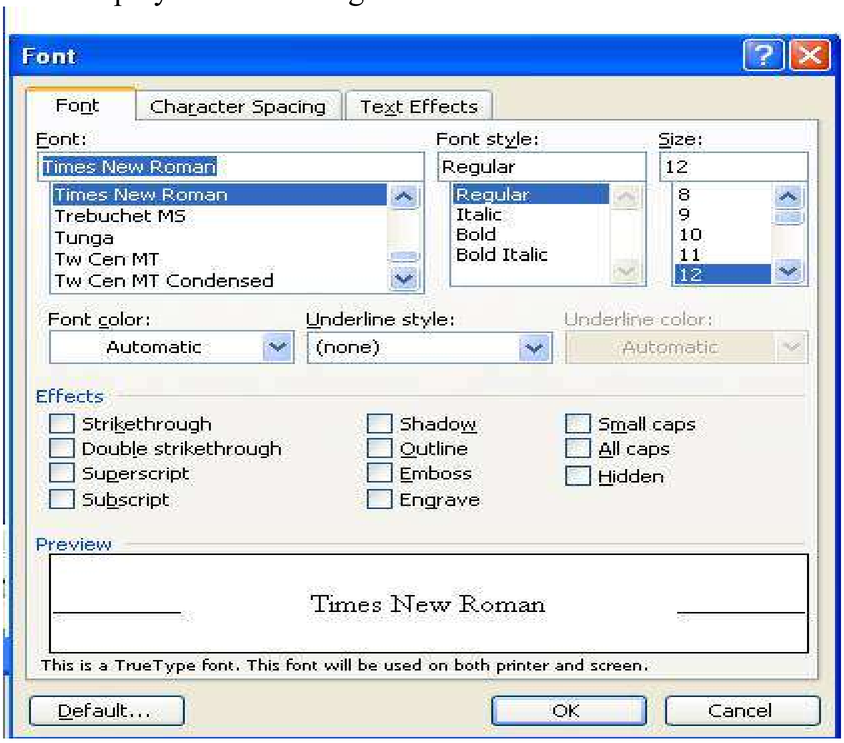
Editing a Document: editing a document generally involves the operations such as selecting the text, moving and copying the text.

Formating a Document:

- Formatting generally refers to the methods of changing the layout and the design of the text according to the requirements.
- Some of the options that can be used to format a document are as follows
    1. Font
    2. Paragraph
    3. Bullets and Numbering.

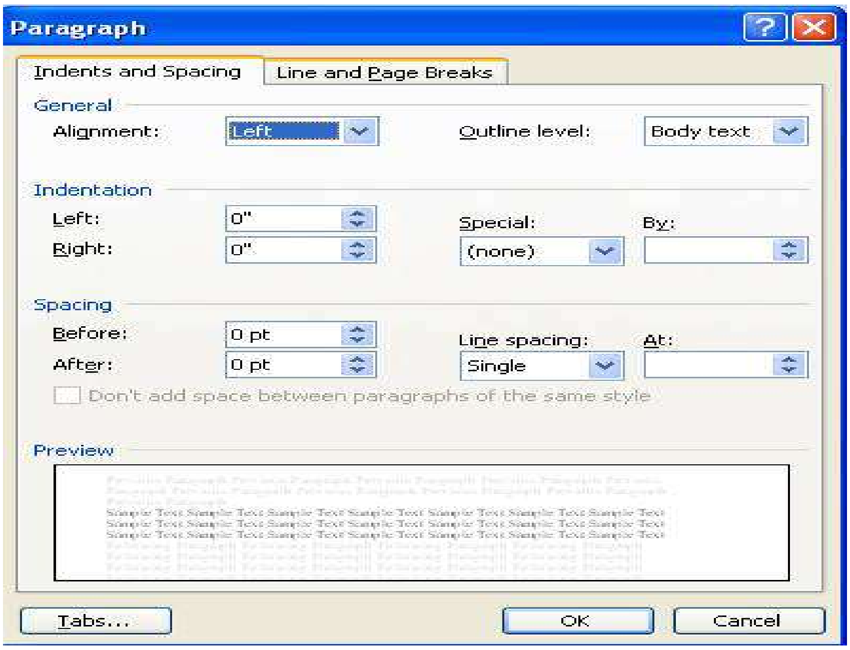
Fonts:

Format□ Font, it will display the font dialog box



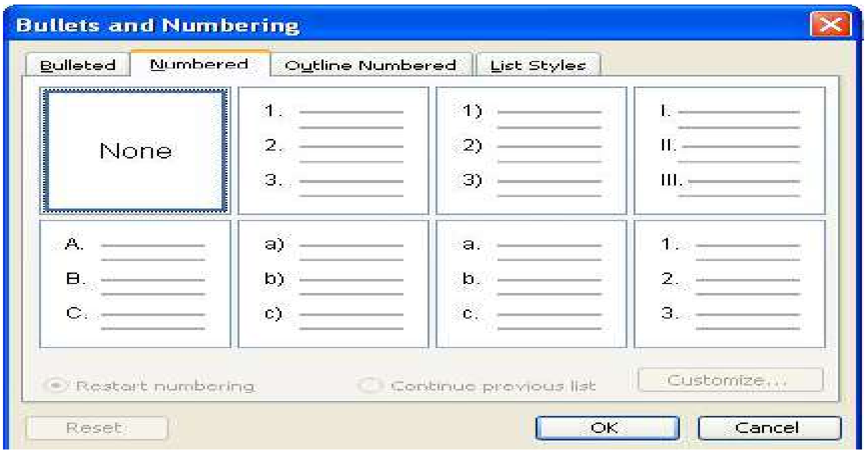
Paragraph:

Format Paragraph



Bullets and numbering:

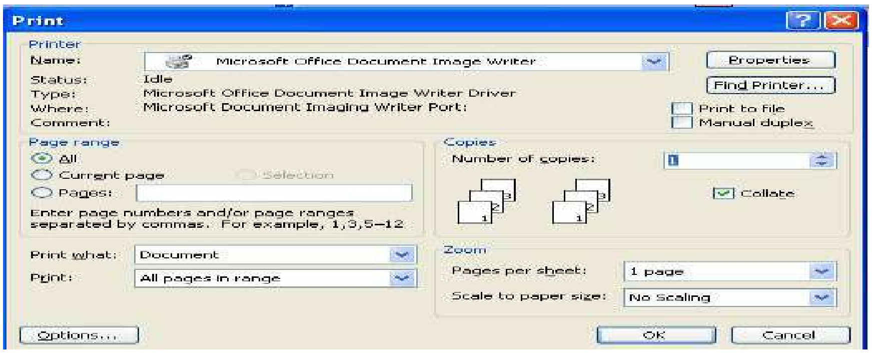
Format Bullets and numbering



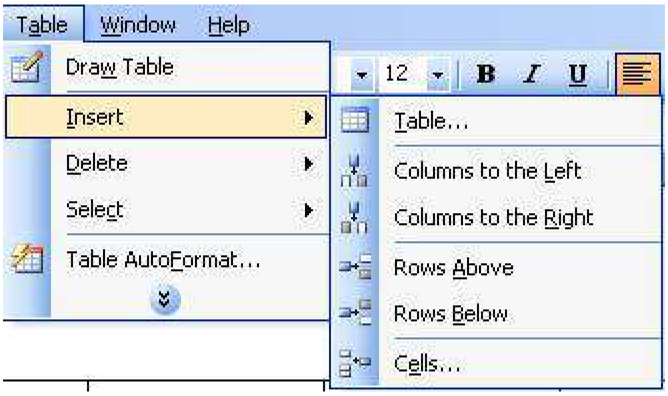
Printing a document:

File Print Preview

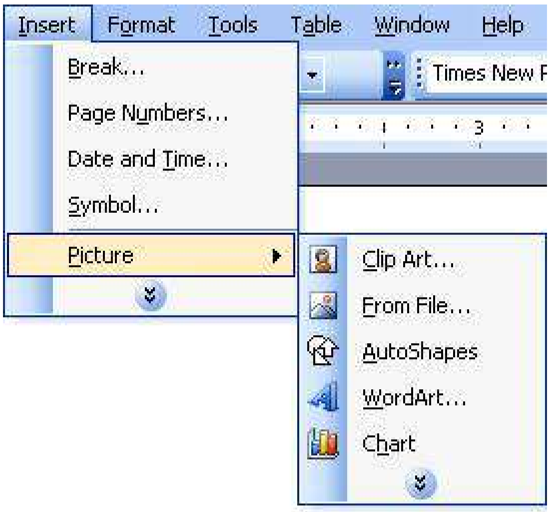
File print



Working with tables:



Adding Picture:



1.16 MAIL MERGE: Univ. Ques. May 2014, Jan 2014, Jan 2013

Univ. Ques. May 2014, Jan 2014, Jan 2013

A mail merge is a method of taking data from a database, spreadsheet, or other form of structured data, and inserting it into documents such as letters, mailing labels, and name tags. It usually requires two files, one storing the variable data to be inserted, and the other containing the information that will be the same for each result of the mail merge and the instructions for formatting the variable data.

Preparing and Selecting the Data Source

The data source is the file where the addresses and other information is kept. Step 3 of the mail-merge procedure calls for you to name your data source, and before you name it, make sure that it is in good working order. The data source can be any number of things, as mentioned above. Word also offers a special dialog box for creating a data source from scratch and storing it in Microsoft Access.

Caution: After you create the data source, do not move it to a different folder or rearrange the merge fields/columns. If you do so, the document can't read the data source correctly and you can't complete the mail-merge.

Note: A field is one category of information (like last name, or zip code). A record comprises all the data about one person or thing. The field is the heading of a table, while records are all of the data underneath.

Generating Form Letters for Mass-Mailings

After you have prepared the data source, you are ready to write and generate form letters. A form letter is a nearly identical letter sent to numerous people. Only the particulars of each recipient are different – the recipient's name, the recipient's address, and perhaps one or two identifying facts about the recipients.

Choose Tools | Letters and Mailings | Mail Merge Wizard and , clicking the Next button as you go along, follow these steps in the Mail Merge task pane to generate form letters for mass-mailings.

Step 1: Select Document Type: Under Select Document Type, select the Letters option button and click the Next hyperlink.

Step 2: Select a Starting Document: Tell Word whether you're starting from scratch or using a letter you've already written for the form letter:

Starting from Scratch: Make sure the Use the Current Document or Start form Template option is selected in the task pane. To write your form letter with the help of a template, click the Select Template hyperlink and choose a template in the Select Template dialog box. The Mail Merge tab offers templates designed especially for mail-merges.

Using a Letter You've Already Written: If you have already written the text of the form letter, select the Start from Existing Document option button, click the Open button, and select the letter in the Open dialog box.

Step 3: Select Recipients: Choose and option button under Select Recipients to choose the data source for the form letters. You see the Mail Merge Recipients dialog box. Click OK if you want to select all the recipients in the source file.

Step 4: Write Your Letter: Compose the text of the letter if you've not already done so, and then enter the address block, greeting line, and merge fields:

Merge fields: Place the cursor where you want a merge field to go, click More Items on the task pane or the Insert Merge Fields button on the Mail Merge toolbar, and double-click the name of a merge field in the Insert Merge Field dialog box. You can also select a field and click the Insert button.

Address block: Place the cursor near or at the top of the letter and click the Insert Address Block button, or click the Address Block hyperlink in the Mail Merge task pane. The Insert Address Block dialog box appears. Enter the address block and click OK.

Greeting line: Place the cursor where the salutation goes and click the Insert Greeting Line button, or click the Greeting Line hyperlink on the Mail Merge toolbar. In the Greeting Line dialog box, fashion a salutation and click OK. Be careful when entering clank spaces and punctuation marks around the merge fields. You can always click the View Merged Data button on the Mail Merge toolbar to see whether punctuation marks and blank spaces will appear correctly.

Step 5: Preview Your Document: Step 5 is your chance to look over the form letters before you print them or create a file for them. Click the Record buttons to see what your document will look like after the mailmerge is complete.

Step 6: Complete the Merge: Now you're ready to go.

Save the Mail Merge in a New Document: You get a new document that you can edit or print another day. Click the Merge to New Document button on the Mail Merge toolbar.

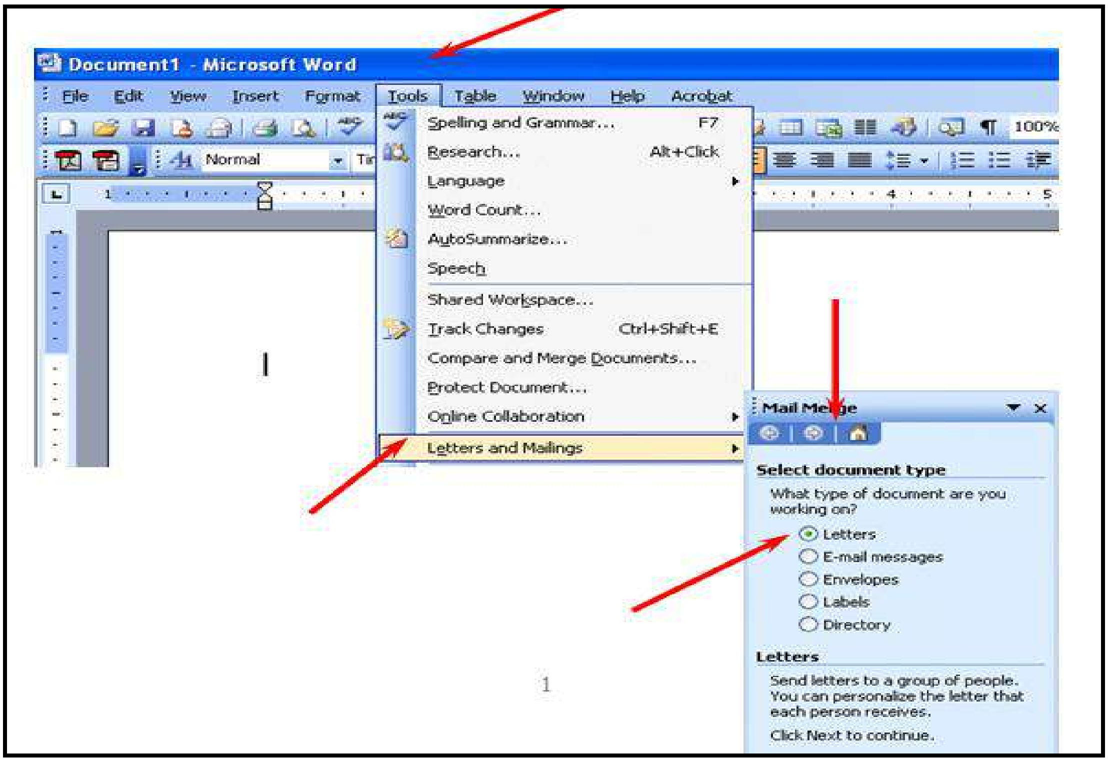
Print the Mail-Merge Right Away: Click the Merge to Printer button on the Mail Merge toolbar.

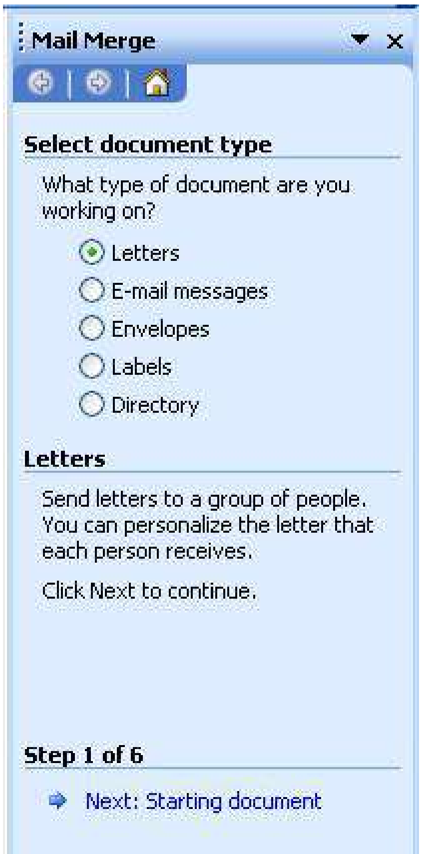
The advantages of using mail merge are:

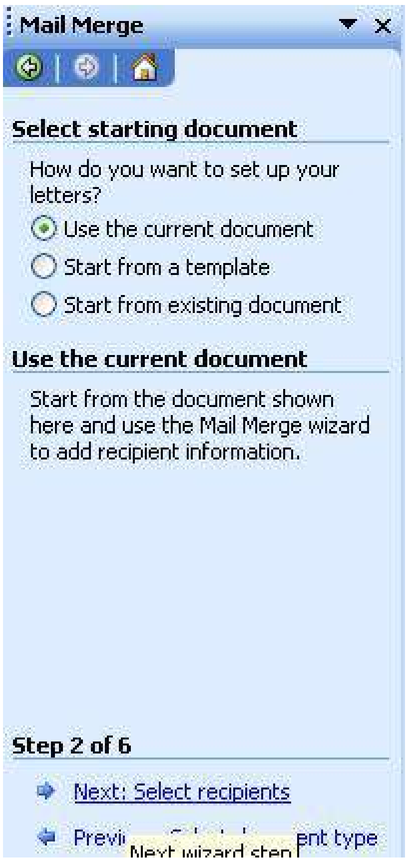
1. Only one document needs to be composed for communicating to an extensive list of interested people, clients or customers.
2. Each document can be personalised i.e. it appears to be have been written specifically to each recipient. It contains details only relevant to the receiver.
3. Many document formats can be developed to use with one database.
4. Errors in transcribing details from one document to another are eliminated. This advantage, of course, depends upon the accuracy of data entry into individual records in the first place.

Using Mail Merge:

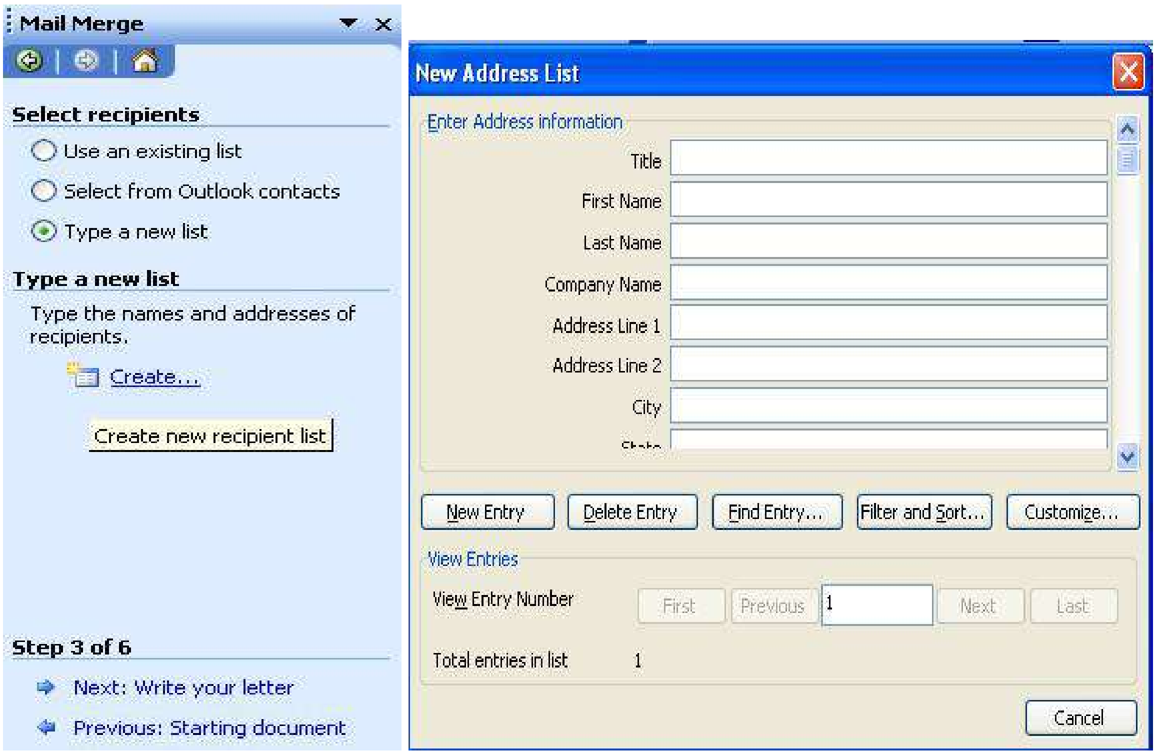
Tools Letter and Mailing Mail Merge Wizard

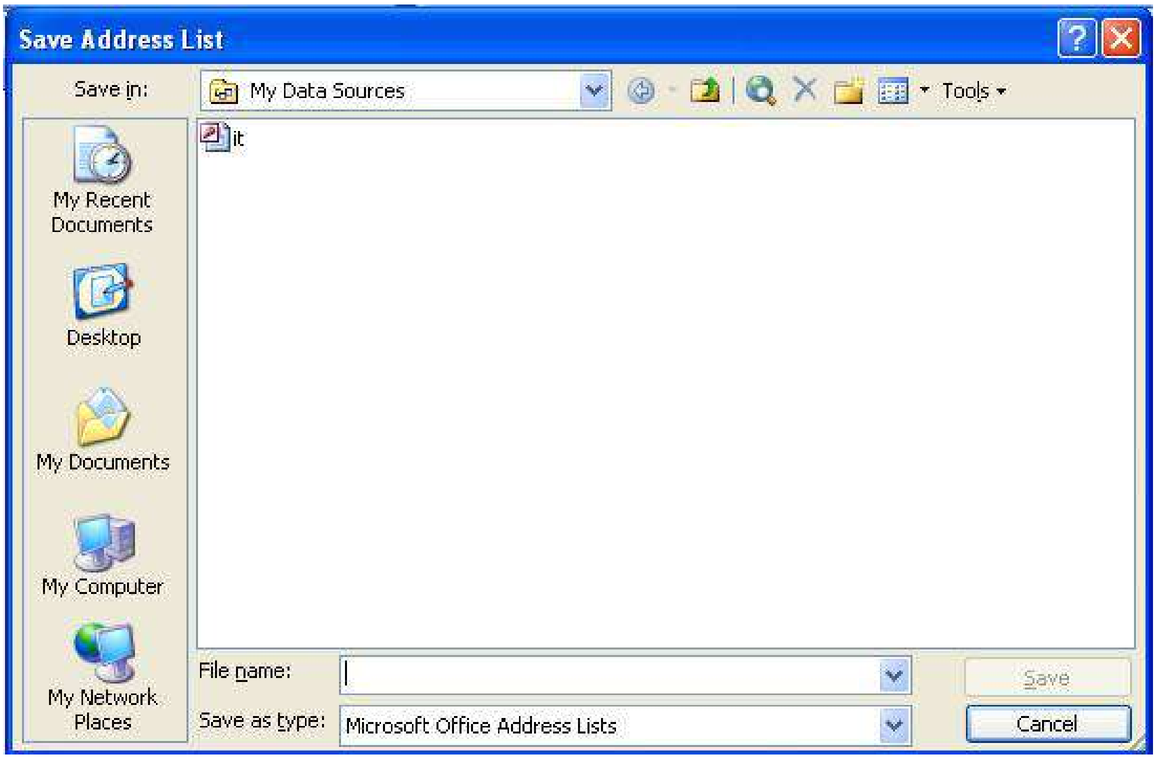

Document1 - Microsoft Word
File Edit View Insert Format Tools Table Window Help Acrobat
Spelling and Grammar... F7
Research... Alt+Click
Language
Word Count...
AutoSummarize...
Speech
Shared Workspace...
Track Changes Ctrl+Shift+E
Compare and Merge Documents...
Protect Document...
Online Collaboration
Letters and Mailings
Mail Merge
Select document type
What type of document are you working on?
Letters
E-mail messages
Envelopes
Labels
Directory
Letters
Send letters to a group of people. You can personalize the letter that each person receives.
Click Next to continue.

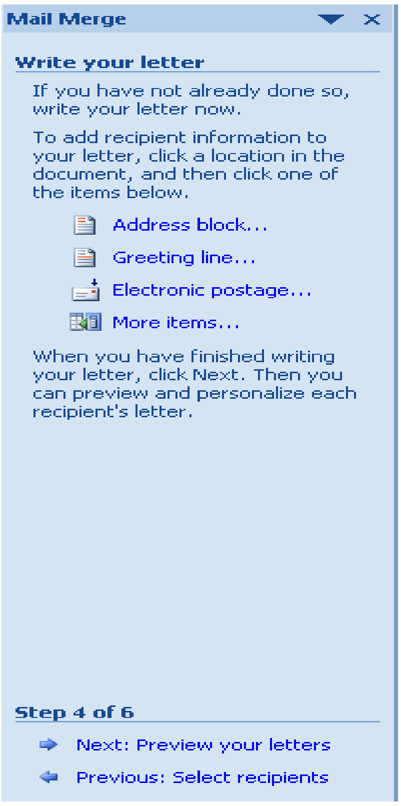

Mail Merge
Select document type
What type of document are you working on?
Letters
E-mail messages
Envelopes
Labels
Directory
Letters
Send letters to a group of people. You can personalize the letter that each person receives.
Click Next to continue.
Step 1 of 6
Next: Starting document

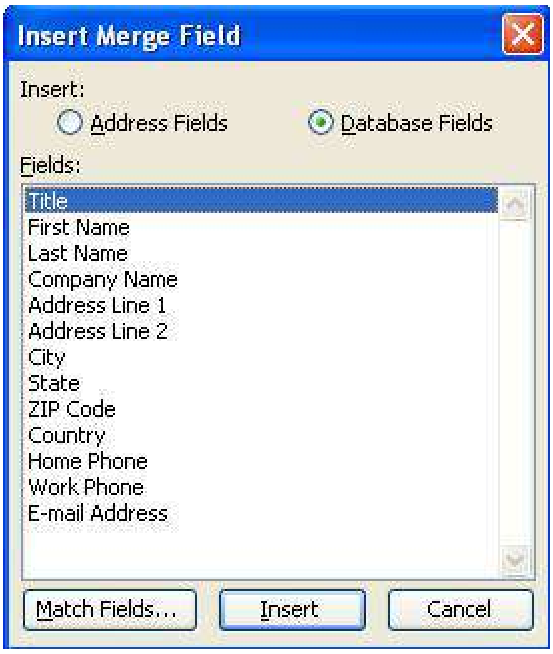

Mail Merge
Select starting document
How do you want to set up your letters?
Use the current document
Start from a template
Start from existing document
Use the current document
Start from the document shown here and use the Mail Merge wizard to add recipient information.
Step 2 of 6
Next: Select recipients
Next wizard step

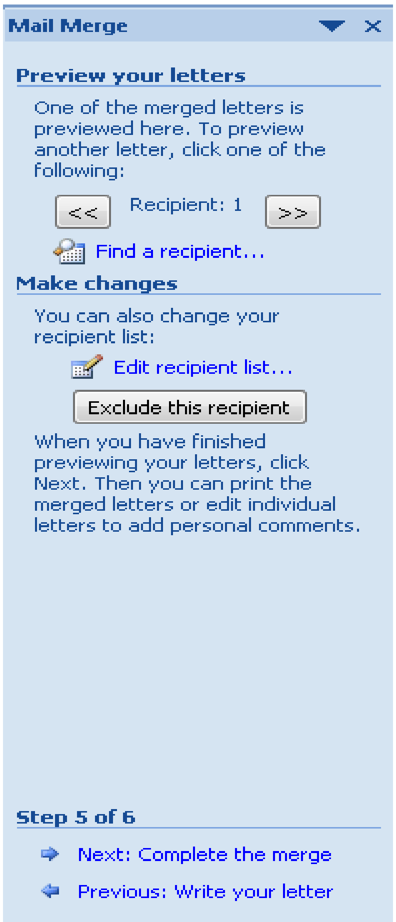
Selecting your merge file (Access, Excel, Word, Outlook)

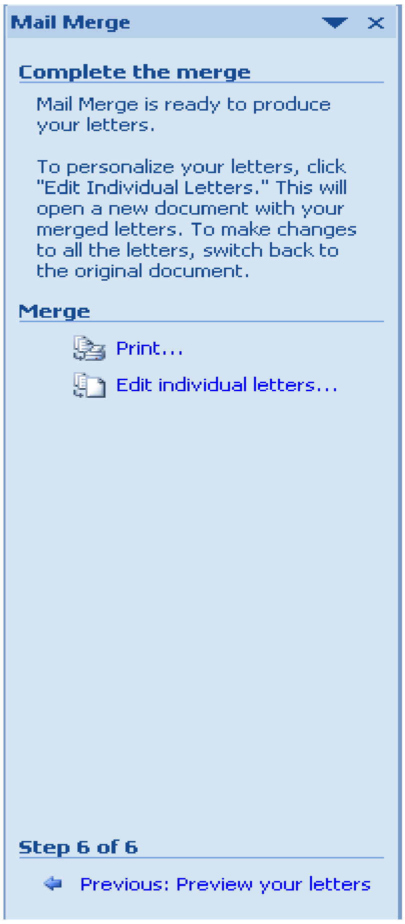





Mail Merge
Write your letter
If you have not already done so, write your letter now.
To add recipient information to your letter, click a location in the document, and then click one of the items below.
Address block...
Greeting line...
Electronic postage...
More items...
When you have finished writing your letter, click Next. Then you can preview and personalize each recipient's letter.
Step 4 of 6
Next: Preview your letters
Previous: Select recipients


Insert Merge Field
Insert:
Address Fields
Database Fields
Fields:
Title
First Name
Last Name
Company Name
Address Line 1
Address Line 2
City
State
ZIP Code
Country
Home Phone
Work Phone
E-mail Address
Match Fields...
Insert
Cancel


Mail Merge
Preview your letters
One of the merged letters is previewed here. To preview another letter, click one of the following:
<<
Recipient: 1
>>
Find a recipient...
Make changes
You can also change your recipient list:
Edit recipient list...
Exclude this recipient
When you have finished previewing your letters, click Next. Then you can print the merged letters or edit individual letters to add personal comments.
Step 5 of 6
Next: Complete the merge
Previous: Write your letter


Mail Merge
Complete the merge
Mail Merge is ready to produce your letters.
To personalize your letters, click "Edit Individual Letters." This will open a new document with your merged letters. To make changes to all the letters, switch back to the original document.
Merge
Print...
Edit individual letters...
Step 6 of 6
Previous: Preview your letters

1.17 SPREAD SHEETS:

Univ. Ques. May 2016

- Spreadsheet are also called *worksheets.*
- Spreadsheets were the earliest application software available for microcomputers.
- A spreadsheet program can be a standalone package or it can be part of an integrated package such as Microsoft Works.

Teachers use spreadsheets in grade books or grade keeping packages to store and calculate grades.

Spreadsheet Features and Capabilities

- Calculations and comparisons: Spreadsheets can calculate and manipulate stored numbers in lots of ways by using different formulas.
- Automatic recalculation.
- Copying cells.
- Placing information in column-row formats for easy reading and interpretation.
- Creating graphs that correspond to data.
- Using worksheets prepared with other programs.

Advantages of Spreadsheets

- Time savings.
- Creating charts.
- Answering "what if" questions.
- Motivation.

Applications of Spreadsheets

Teacher Productivity:

- Gradekeeping.
- Club and/or classroom budgets.
- Computerized checkbooks for clubs or other organizations.
- Attendance charts.
- Performance assessment checklist.
- Class inventory.

Teaching and Learning Activities

- Demonstrations.
- Student products.
- Problem solving activities.
- Storing and analyzing data.
- Projecting grades.

Common Mistakes and Misconceptions:

- Forgetting to highlight cells to be formatted.
- Difficulty developing formulas and using predefined functions

Shortcuts for Spreadsheet:

EXCEL SHORTCUT KEYS

See our Microsoft Excel page for additional help and information.

Shortcut Keys	Description
F2	Edit the selected cell.
F5	Go to a specific cell. For example, C6.
F7	Spell check selected text and/or document.
F11	Create chart.
Ctrl + Shift + ;	Enter the current time.
Ctrl + ;	Enter the current date.
Alt + Shift + F1	Insert New Worksheet.
Shift + F3	Open the Excel formula window.
Shift + F5	Bring up search box.
Ctrl + A	Select all contents of the worksheet.
Ctrl + B	Bold highlighted selection.
Ctrl + I	Italic highlighted selection.
Ctrl + K	Insert link.
Ctrl + U	Underline highlighted selection.
Ctrl + 5	Strikethrough highlighted selection.
Ctrl + P	Bring up the print dialog box to begin printing.
Ctrl + Z	Undo last action.
Ctrl + F9	Minimize current window.
Ctrl + F10	Maximize currently selected window.
Ctrl + F6	Switch between open workbooks / windows.
Ctrl + Page up	Move between Excel work sheets in the same Excel document.

PREPARATION OF WORKSHEETS:

- Ms Excel is an application that allows us to **create spreadsheet**, which is represented in the form of a table containing rows and columns.
- The horizontal sequence is called row, vertical sequence is called column.
- A row is identified by its **row header** (1, 2 , 3….)
- A column is identified by its **column header** (A, B, C….)
- Intersection of rows and column is called **cell**
- We can perform various mathematical operations using formulas, such as addition, subtraction, multiplication and division

- Saved with the extension **.xls**
- We can insert graph, images …..

Accessing MS Excel:

1 Using Menu:

Start□ Programs□Microsoft Office□ Microsoft Office Excel 2003

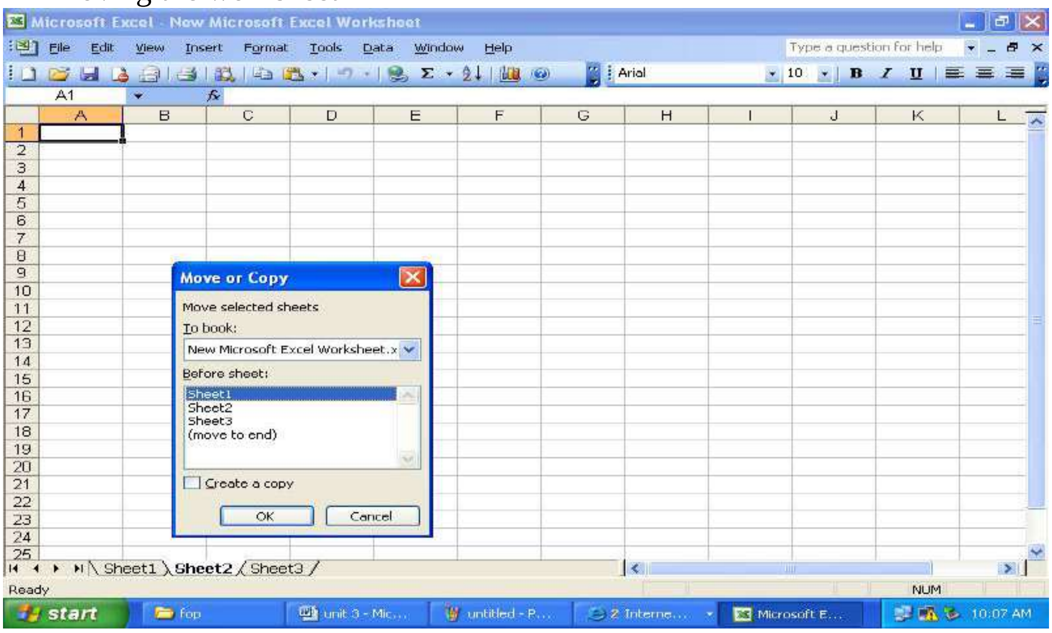
2 Run command

Start□ Run to display the Run dialog box
Type **Excel** in the open text box and click OK

Basic operations performed in MS Excel:

- Creating
- Saving
- Modifying
- Renaming
- Deleting
- Moving
- Editing

Moving the worksheet



Using formulas and functions:

- Formulas are equations that are used for manipulating the data of the worksheet.
- To write a formula in a cell, first insert the equal to sign (=), then enter the value

- Formula consists of following parts:
  1. Constants: numbers eg: 20, 30
  2. Operators: +, -, *, /

3. Functions: IF, SUM, COUNT
4. References: Eg: =(E12)

- Operators can be categorized into different types

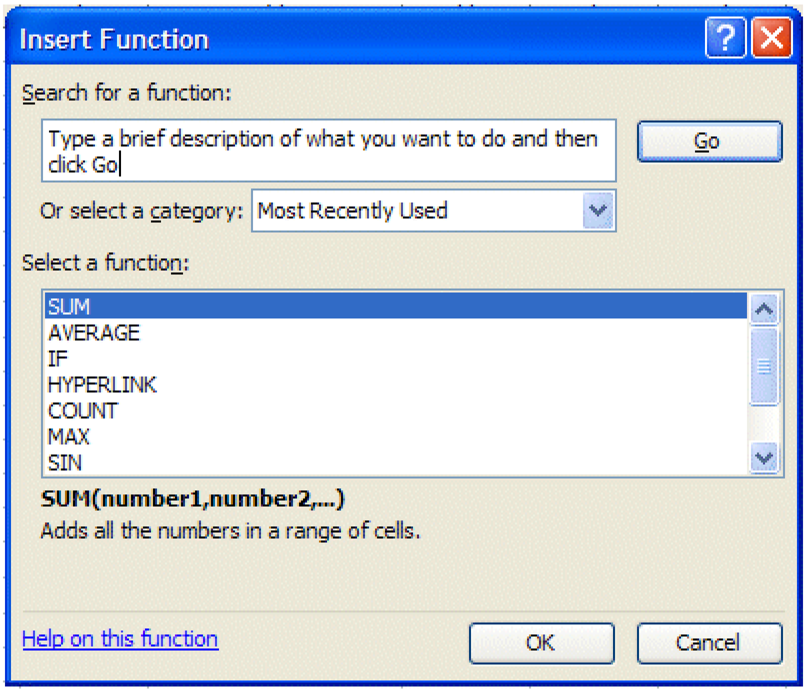
1. Arithmetic : +,-, *,/, % and ^
2. Text Operator: & (ampersand- concatenate two strings)
3. Comparison: <,>,<=,>=,<>
4. Range: = SUM(A5 **:** A14)
5. Union = SUM (A8**,** B10**,** D16)
6. Intersect =SUM (D14:D25 B12: B23)

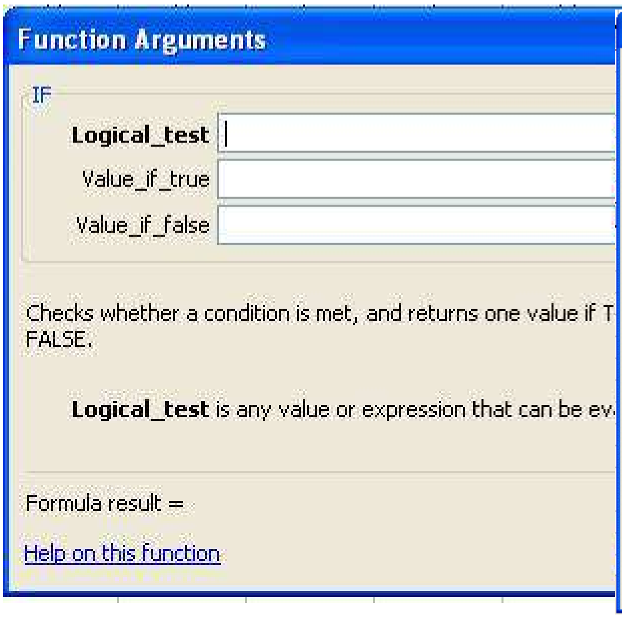
- Different functions used

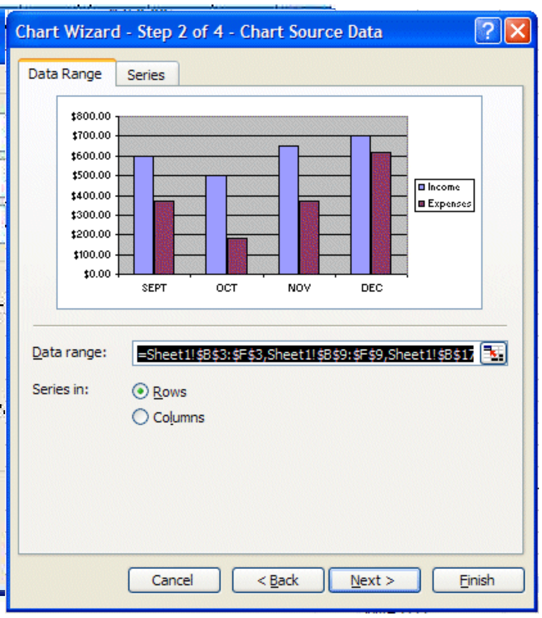
1. Financial: FV, PMT, RATE
2. Math & Trig : SIN,COS, EXP, FACT
3. Date & Time: DATE, DAY, MONTH, TIME, YEAR
4. Text: CHAR, CONCATENATE, FIND, LEN.
5. Logical: AND, IF, NOT, OR

- **To insert formulas:**

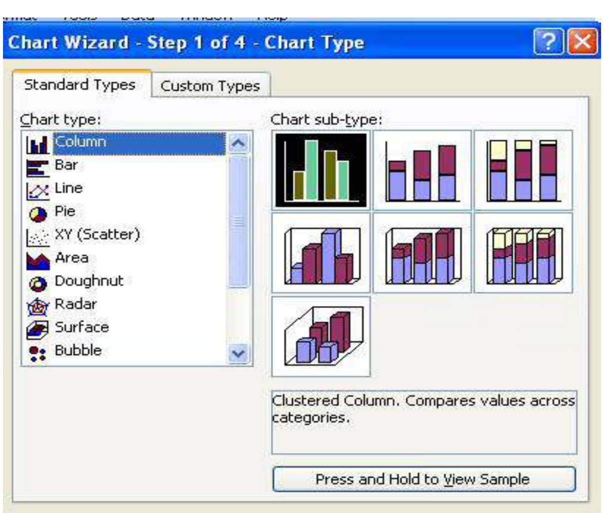
Insert□ Function

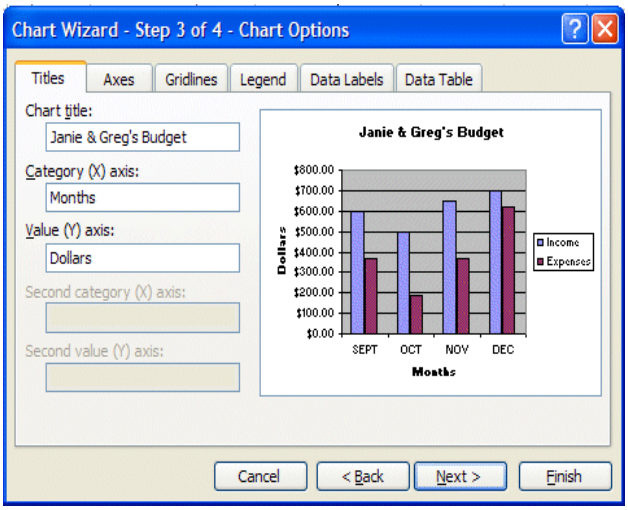






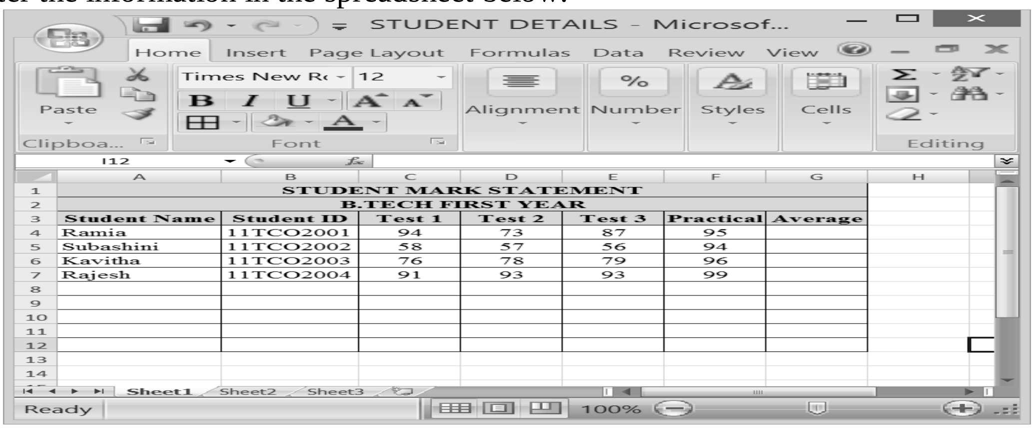
Creating chart:





PREPARATION OF WORKSHEET FOR STUDENTS MARK DETAILS

1. Enter the information in the spreadsheet below.

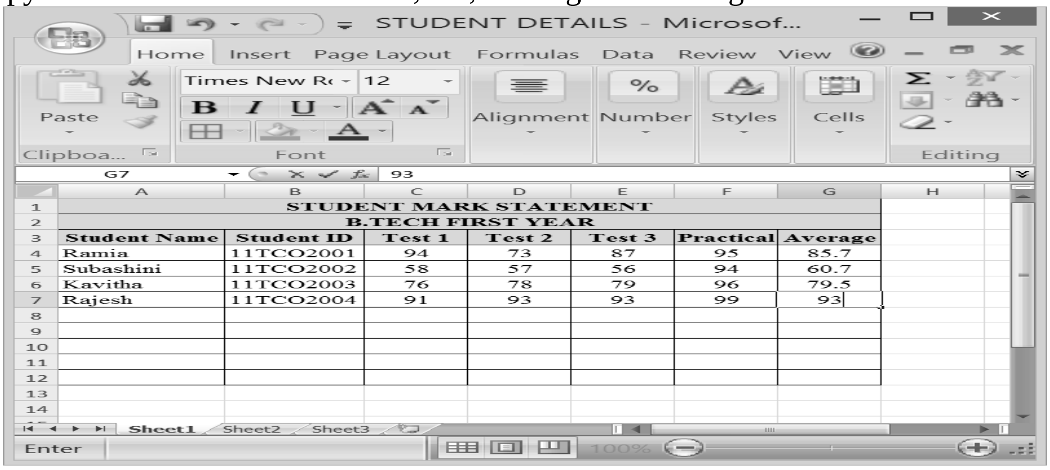


STUDENT MARK STATEMENT						
B.TECH FIRST YEAR						
Student Name	Student ID	Test 1	Test 2	Test 3	Practical	Average
Ramia	11TCO2001	94	73	87	95	
Subashini	11TCO2002	58	57	56	94	
Kavitha	11TCO2003	76	78	79	96	
Rajesh	11TCO2004	91	93	93	99	

2. Enter the formula in the cell G to find the average

G4 = C4*.3 + D4*.3 + E4*.3 + F4*.1

3. Copy the cell G5 in to the cells G5, G6, G7 to get the averages.



4. Enter the information's given below

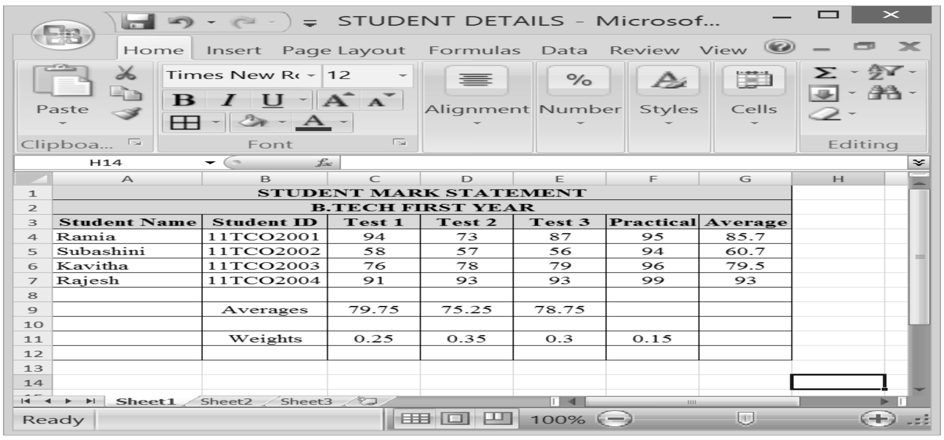
B9: Averages
C9 = (C4+C5+C6+C7)/4
D9 = SUM (D4:D7)/4
E9 = AVERAGE (E4:E7)

5. Enter the information given below

B11: Weights
C11: .3
D11: .3
E11: .3
F11: .3

6. Enter the formula in G4 and copy it to G5, G6, G7

G4: =$C$11*C4 + =$D$11*D4 + =$E$11*E4 + =$F$11*F4

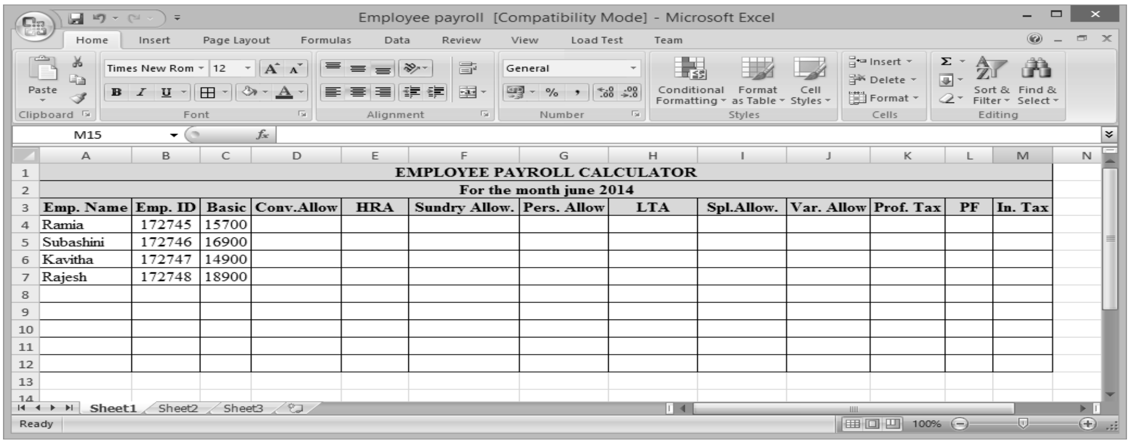


7. Save the above worksheet as worksheet_name.xls and exit.

WORKSHEET FOR EMPLOYEE PAYROLL

Univ. Ques. May 2013

1. Enter the information in the spreadsheet given below.



EMPLOYEE PAYROLL CALCULATOR

For the month june 2014

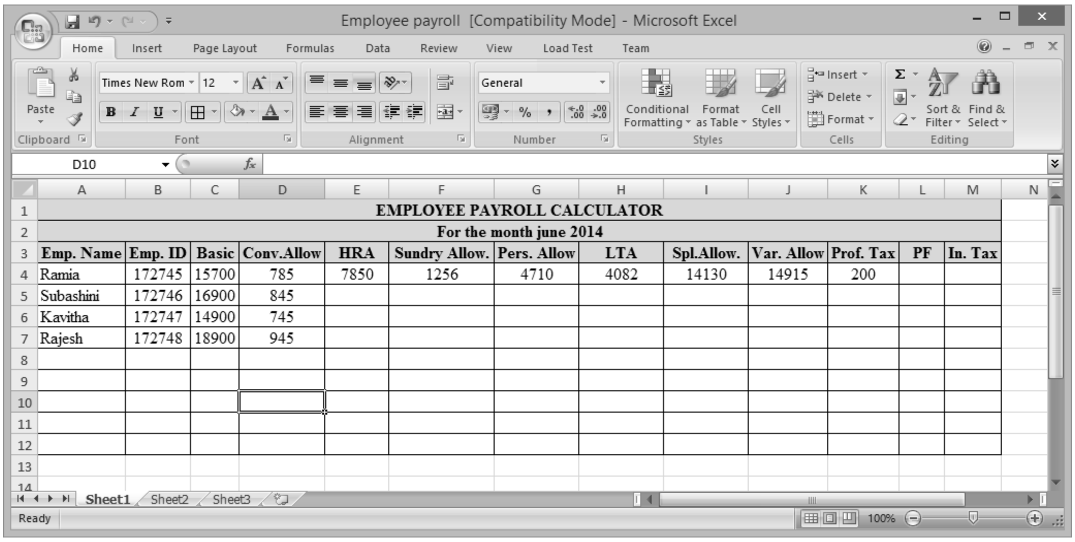
Emp. Name	Emp. ID	Basic	Conv.Allow	HRA	Sundry Allow.	Pers. Allow	LTA	Spl.Allow.	Var. Allow	Prof. Tax	PF	In. Tax
Ramia	172745	15700										
Subashini	172746	16900										
Kavitha	172747	14900										
Rajesh	172748	18900										

2. Enter the basic pay of the employee and remaining details can be calculated using the formulas in spreadsheet.

3. Calculate the details specified herewith with the following percentages

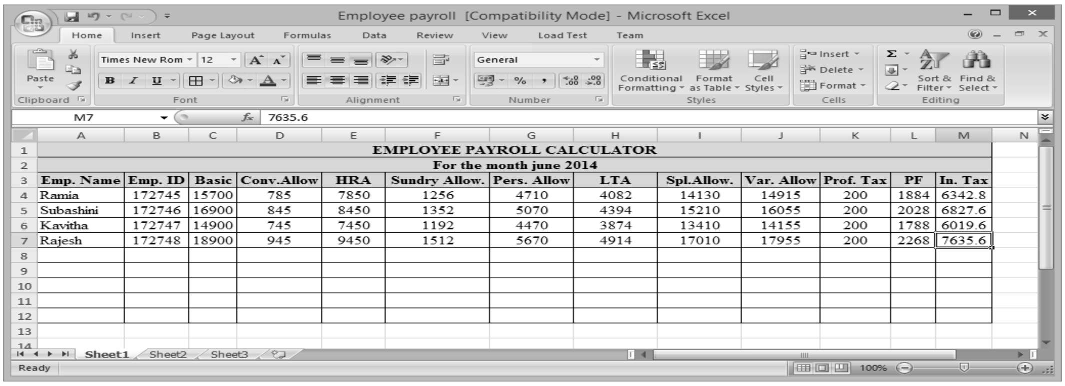
   a. Conveyance Allowances - 5% of Basic Pay
   b. House Rent Allowances(HRA) - 50% of Basic Pay
   c. Sundry Allowances - 8% of Basic Pay
   d. Personal Allowances - 30% of Basic Pay
   e. Leave Travelling Allowances - 26% of Basic Pay
   f. Special Allowances - 90% of Basic Pay
   g. Variable Allowances - 95% of Basic Pay
   h. Professional Tax - Rs. 200
   i. Provident Fund(PF) - 12% of Basic Pay
   j. Income Tax (In. Tax) - 10% per month of total earning per hour

4. To calculate the details as per specified percentages above, use the following formulas.
   a. D4 = C4*5% Where C4 is the Basic Pay
   b. E4 = C4*50%
   c. C4 = C4*8%
   d. F4 = C4*30%
   e. G4 = C4*26%
   f. H4 = C4*90%
   g. I4 = C4*95%
   h. K4 = 200
   i. L4 = C4*12%
   j. M4 = (SUM(C4:J4)*12)*10%/12



EMPLOYEE PAYROLL CALCULATOR												
For the month june 2014												
Emp. Name	Emp. ID	Basic	Conv.Allow	HRA	Sundry Allow.	Pers. Allow	LTA	Spl.Allow.	Var. Allow	Prof. Tax	PF	In. Tax
Ramia	172745	15700	785	7850	1256	4710	4082	14130	14915	200		
Subashini	172746	16900	845									
Kavitha	172747	14900	745									
Rajesh	172748	18900	945									

5. Now copy the formulas to cells to calculate the details for the other employees. The spreadsheet looks like below.



EMPLOYEE PAYROLL CALCULATOR												
For the month june 2014												
Emp. Name	Emp. ID	Basic	Conv.Allow	HRA	Sundry Allow.	Pers. Allow	LTA	Spl.Allow.	Var. Allow	Prof. Tax	PF	In. Tax
Ramia	172745	15700	785	7850	1256	4710	4082	14130	14915	200	1884	6342.8
Subashini	172746	16900	845	8450	1352	5070	4394	15210	16055	200	2028	6827.6
Kavitha	172747	14900	745	7450	1192	4470	3874	13410	14155	200	1788	6019.6
Rajesh	172748	18900	945	9450	1512	5670	4914	17010	17955	200	2268	7635.6

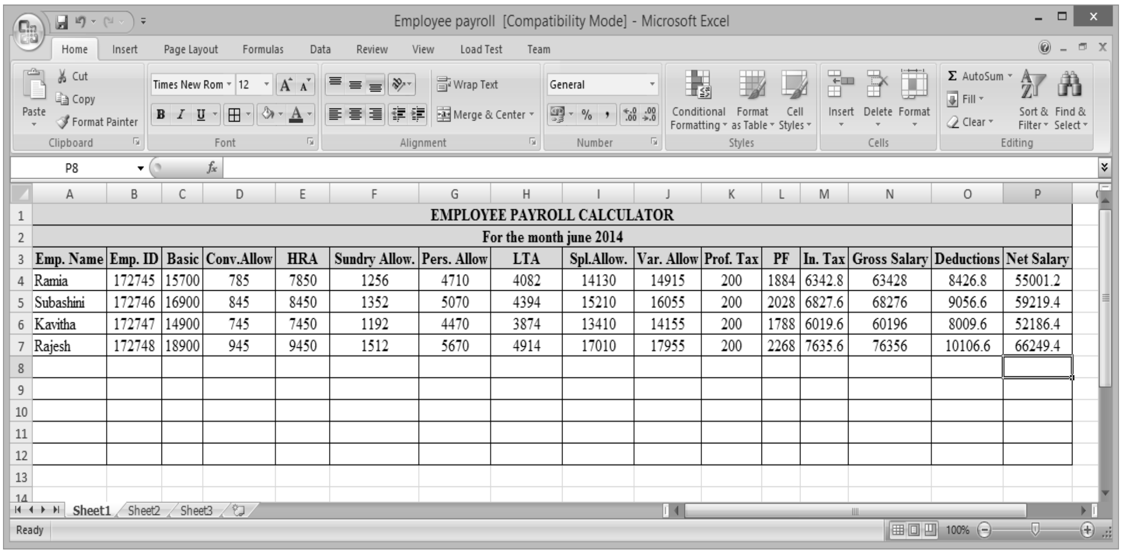
6. Now let us calculate the Gross Salary, Deductions and Net Salary for the employee

Gross Salary = Basic + Conv. Allow. + HRA + Sundry Allow. + Pers. Allow + LTA + Spl. Allow + Var. Allow.
that is
N4 = C4+D4+E4+F4+G4+H4+I4+J4

Deductions = Prof. Tax + PF + In. Tax
that is
O4 = K4+L4+M4

Net Salary: Gross Salary - Deduction
that is
P4 = N4 – O4

7. Enter all the formulas the final worksheet looks like below



EMPLOYEE PAYROLL CALCULATOR															
For the month june 2014															
Emp. Name	**Emp. ID**	**Basic**	**Conv.Allow**	**HRA**	**Sundry Allow.**	**Pers. Allow**	**LTA**	**Spl.Allow.**	**Var. Allow**	**Prof. Tax**	**PF**	**In. Tax**	**Gross Salary**	**Deductions**	**Net Salary**
Ramia	172745	15700	785	7850	1256	4710	4082	14130	14915	200	1884	6342.8	63428	8426.8	55001.2
Subashini	172746	16900	845	8450	1352	5070	4394	15210	16055	200	2028	6827.6	68276	9056.6	59219.4
Kavitha	172747	14900	745	7450	1192	4470	3874	13410	14155	200	1788	6019.6	60196	8009.6	52186.4
Rajesh	172748	18900	945	9450	1512	5670	4914	17010	17955	200	2268	7635.6	76356	10106.6	66249.4

8. Save and exit.